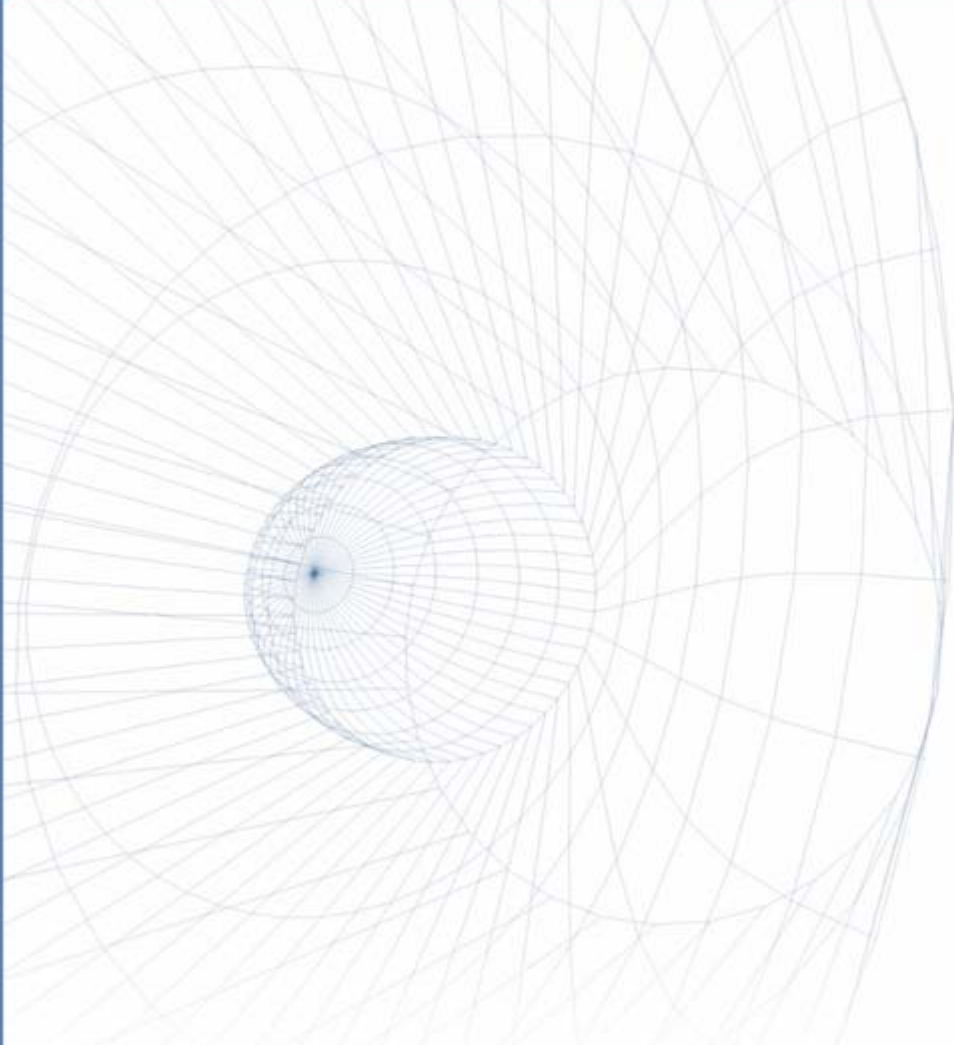




AVEVA

MARINE

Marine Software Installation Guide

www.aveva.com

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Marine Software Installation Guide

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1 Introduction

The Marine Software Installation Guide explains how to install and set up Marine & Associated Products on a workstation running Microsoft Windows. For completely up-to-date information, refer to the AVEVA Website at <http://www.aveva.com/products/support/support.html> and the Customer Support Hardware pages. The Web pages also give hints on ways of setting up and running Marine on different computers in order to achieve the most efficient performance.

1.1 Assumptions

The Marine Software Installation Guide assumes that:

- Users are familiar with the concepts of folder/file hierarchies for storing data on disks and with basic Windows terminology and commands. If not, the user can also refer to the operating system documentation while carrying out the installation.
- Users have already gained access to the Marine installation files via a disk, from the AVEVA Web site, or by some other route.
- Users have Local Administrator rights.

1.2 Guide Structure

The Marine Software Installation Guide is divided into the following sections:

<i>Hardware and Software Requirements</i>	lists the hardware and software requirements.
<i>AVEVA Marine and Associated Products Installation</i>	describes how to install and license the software.
<i>Set Up the Windows Environment</i>	explains how to configure the Windows operating system before running Marine.
<i>Run Marine Under Windows</i>	gives some advice on using Marine in a Windows environment.
<i>Troubleshooting</i>	describes solutions for some of the errors that you could encounter when running Marine.

1.3 Associated Products

There are various AVEVA products that may be used in conjunction with Marine. The user can consider installing these at the same time. Those closely associated with Marine include the following DESIGN products:

- Global and Global WCF Client - for synchronisation of projects between world-wide sites
- Schematic 3D Integrator - Integration application to make sure there is data consistency between P&ID and the 3D model
- Area Based ADP - Automates the production of volumetric drawings using specified templates, thus simplifying their update after a change to the model
- ExPlant-A - DXF export driver
- ExPlant-I - DGN export driver
- Pipe Stress Interface - for exchanging information between the Marine and the CEASAR II pipe stress system
- Initial Design

For details of the installation of these 'add-on' products, refer to the relevant installation guide. If they are required, it may be advisable to read these before installing Marine.

2 Hardware and Software Requirements

The user should refer to the *Systems Requirements* guide and make sure that the system and network meets all of the requirements before attempting to install the software.

2.1 AVEVA Marine 12.1 Software Installation

AVEVA Marine software is supplied on a disk.

The disk contains

- Release Notes
- an installable version of Microsoft's .NET Framework
- AVEVA Licensing System

Insert the disk in your Workstation disk drive, and then follow the instructions.

If the disk does not start, click **Start > Run** and then browse for the **Setup.exe**.

2.1.1 AVEVA License File

To operate, AVEVA software must be able to access the correct license file, which should be installed on either the Server, on your workstation, or on a networked workstation. Refer to the AVEVA Licensing System guide. News and update information is also available on the AVEVA support web-site.

Note: Although the disk contains several AVEVA products, the user will only be able to use the items that are licensed to them. The user will get an on-screen notification to warn them when the license is due for renewal.

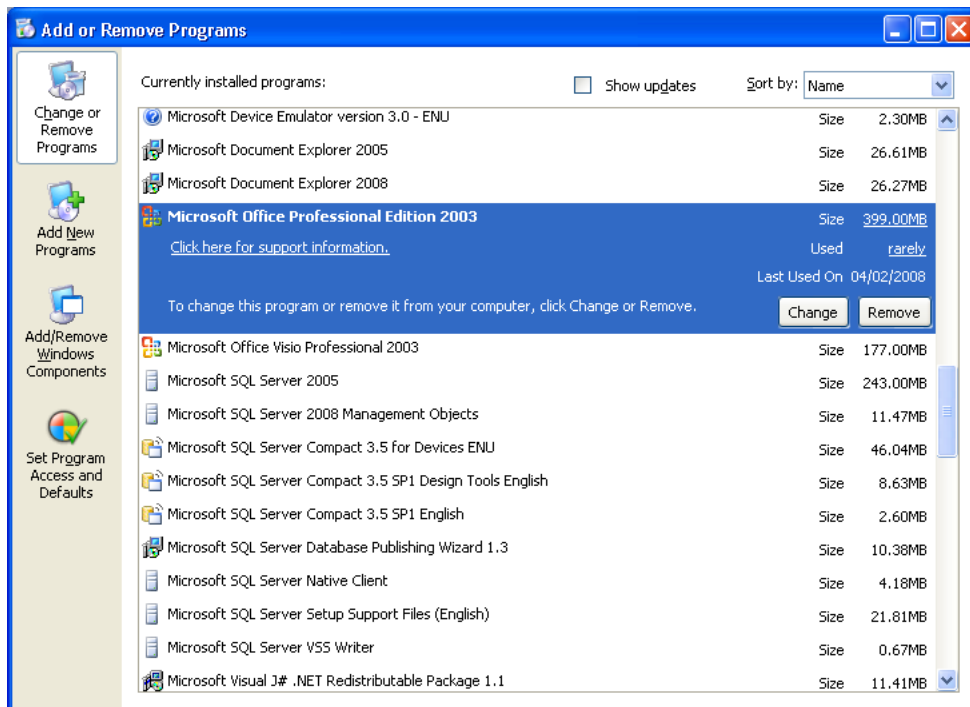
2.1.2 .NET Framework Installation

The graphical user interface requires that Microsoft's .NET Framework be installed. The .NET Framework is supplied on the disk and if it has not been previously installed the installation will install it automatically when the installation is run from the disk.

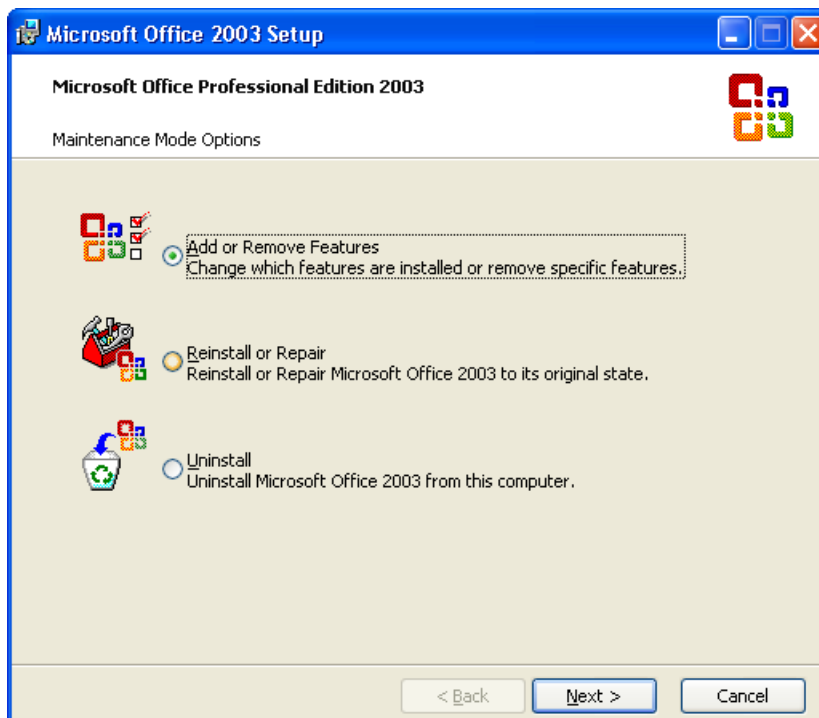
2.2 Enable Microsoft Excel Interop Assembly

Several of AVEVA applications use a facility in the AVEVA GridControl to import an Excel file, for example, Equipment Import, Cable Import, and Diagrams. To make sure that the Excel import facility will work the user must make sure that the Microsoft Excel Interop Assembly is enabled on each machine where it will be used.

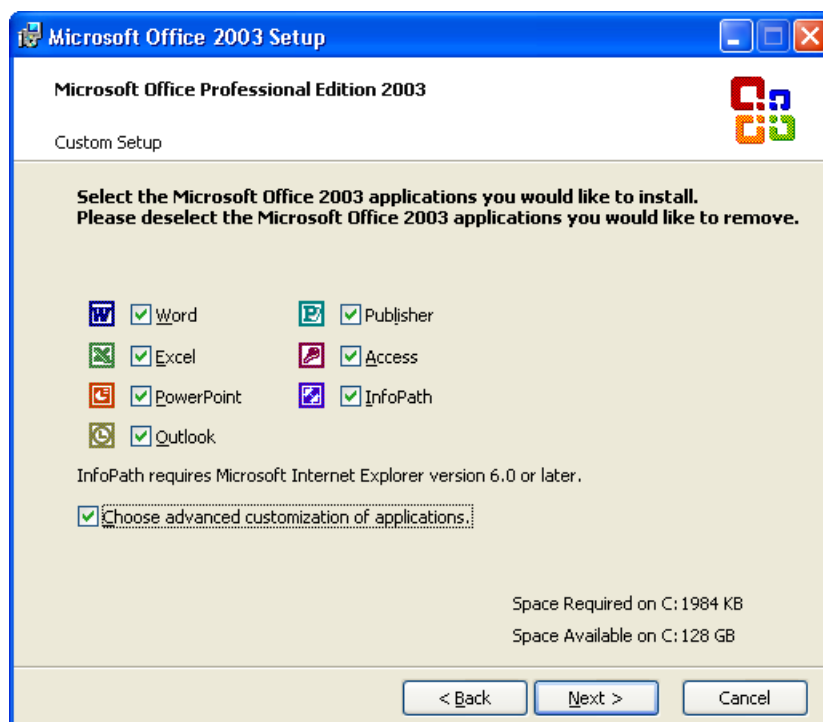
Open the **Add or Remove Programs** utility from the **Control Panel**, and select the version of Microsoft Office being used, the one shown here is **Microsoft Office Professional Edition 2003**.



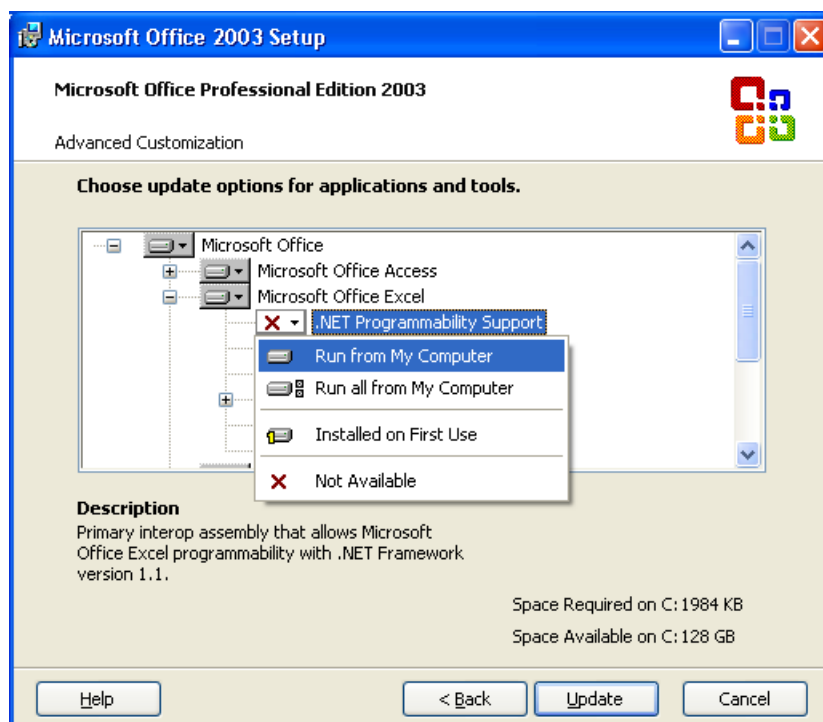
Click **Change** to display the **Maintenance Mode Options**.



Make sure **Add or Remove Features** is checked and click **Next**:



Check the option for **Choose advanced customisation of applications**, and click **Next**.

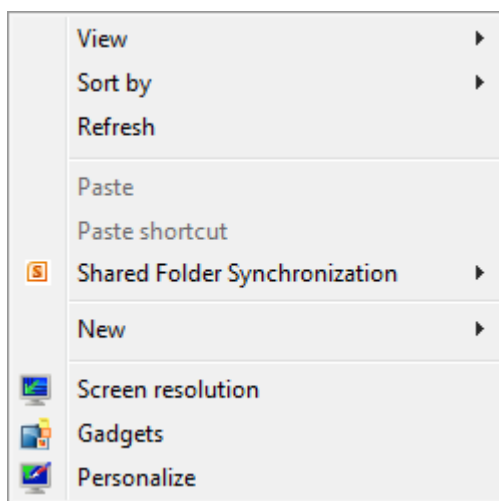


Under the **Microsoft Office Excel** application make sure that **.NET Programmability Support** is enabled. If **.NET Programmability Support** is not enabled then select the option **Run from My Computer**, and click **Update**.

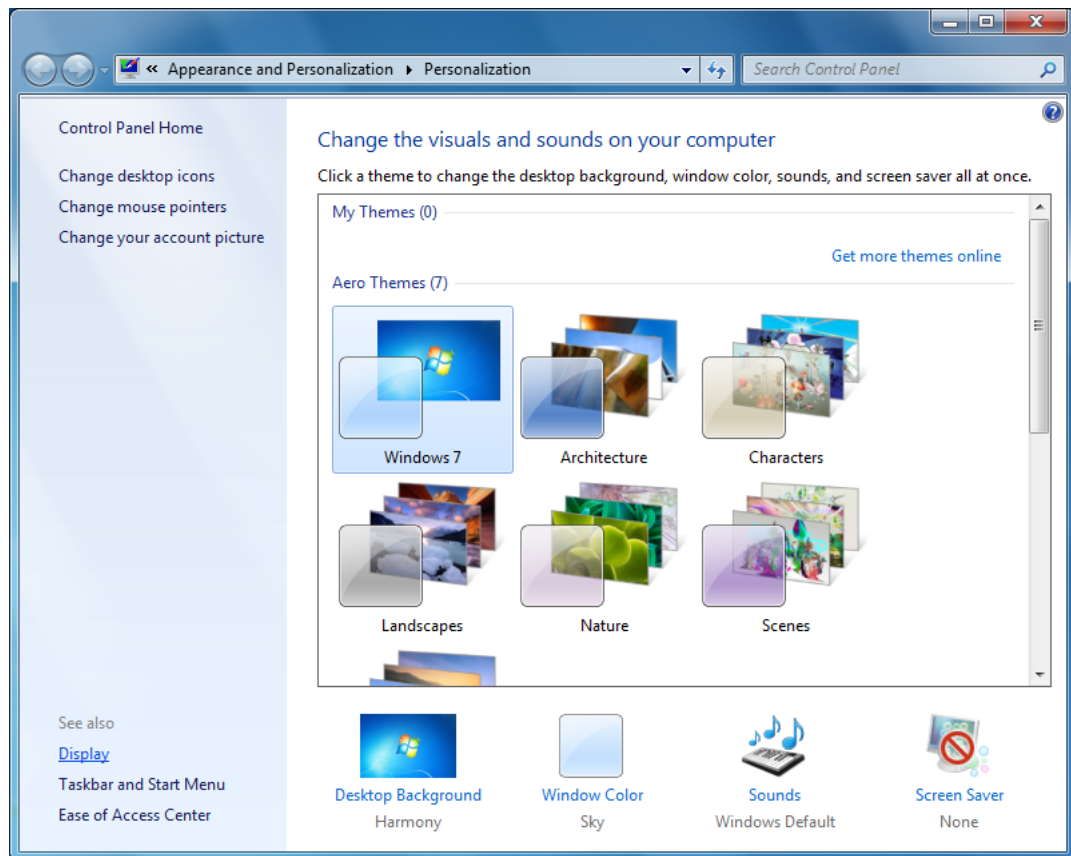
2.3 Font Size

During the installation of Windows 7, the install process decides what it believes to be the best setting for the font size and sets this as the Default. AVEVA only supports the setting **Smaller** and suggests that the user changes their setup to make **Smaller** the default setting.

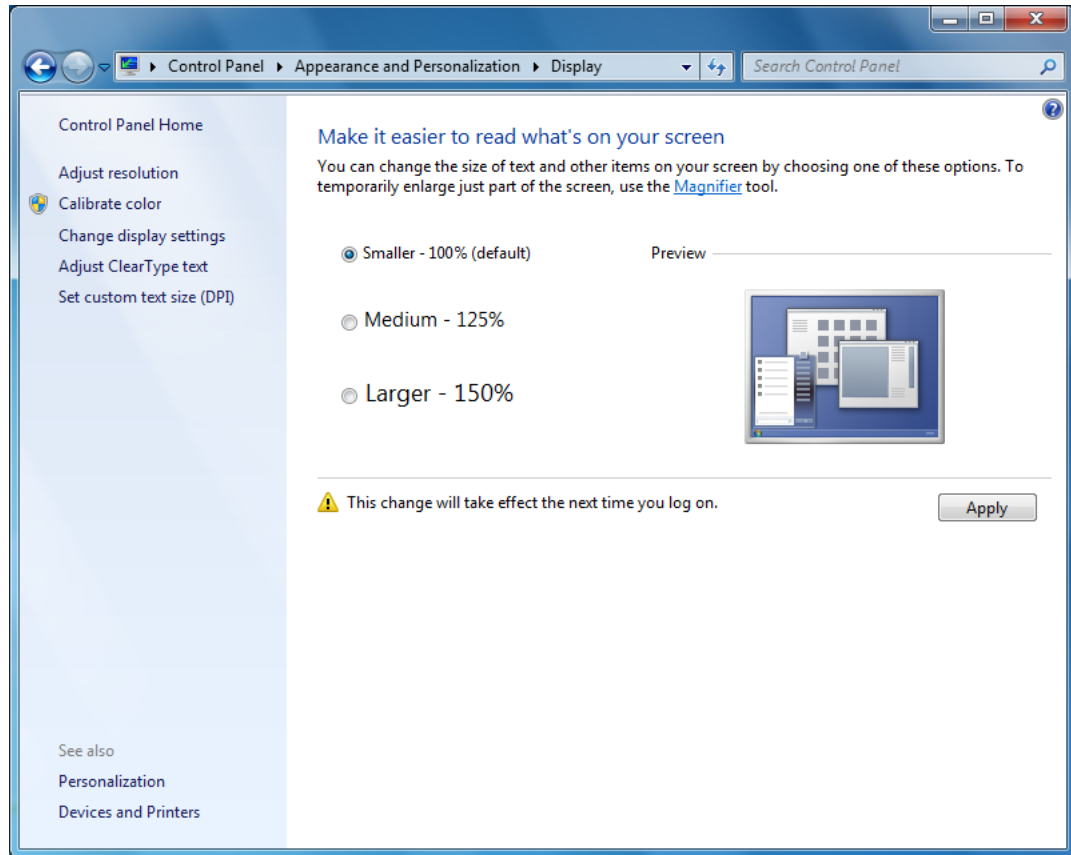
To change the default setting, right-click on the desktop.



Select **Personalize** from the menu to display the control panel.



Click **Display** to display the screen for setting the Font size.



Select **Smaller** and click **Apply**. The user will have to restart the system to save the settings.

3 AVEVA Marine and Associated Products Installation

The AVEVA Marine and Associated Products Installation is supplied on a disk in an installation called Marine. The disk contains an installable version of Microsoft's .NET Framework. Also on the disk is a folder containing the Installation instructions and user documentation, and it is recommended that the user carefully read all of the Installation instructions before installing the software.

It is recommended that the default settings are used for folder paths etc. unless the user has good reasons for doing otherwise.

Important: Do not use spaces in installation folder path names. This includes the use of "Program Files".

Important: After the Marine installation has completed an installation of the AVEVA Database Cache Service is started automatically. A prerequisite for AVEVA Database Cache Service is that the Microsoft Message Queuing (MSMQ) is enabled. For a description of enabling the MSMQ, please refer to chapter [AVEVA Database Cache Service](#).

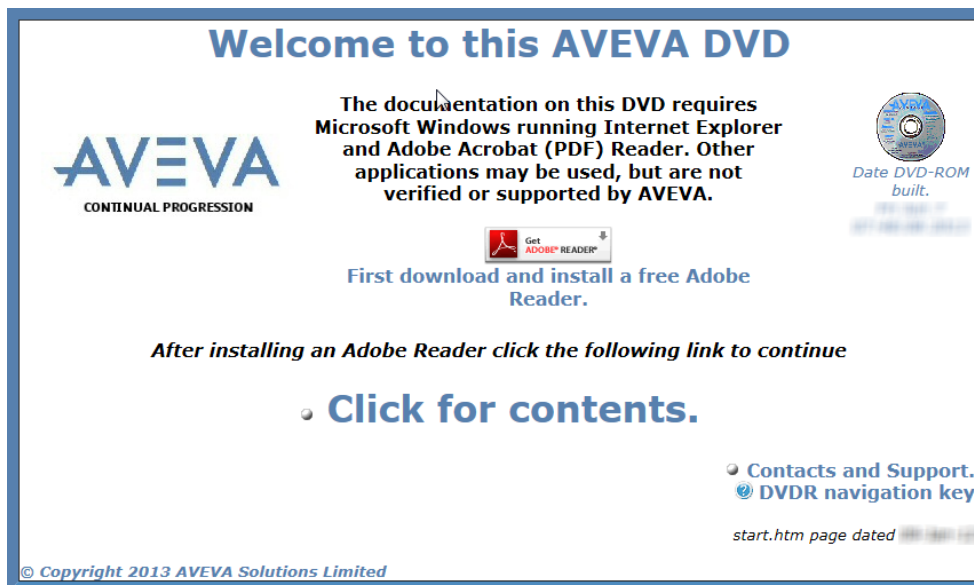
Using the disk to install AVEVA Marine and Associated products, insert the disk and on the Welcome screen click **Click for contents**.

Alternatively the user can click the setup.exe file to install the software manually. Using the setup.exe option bypasses the first three windows and displays the **Marine Setup Wizard** window.

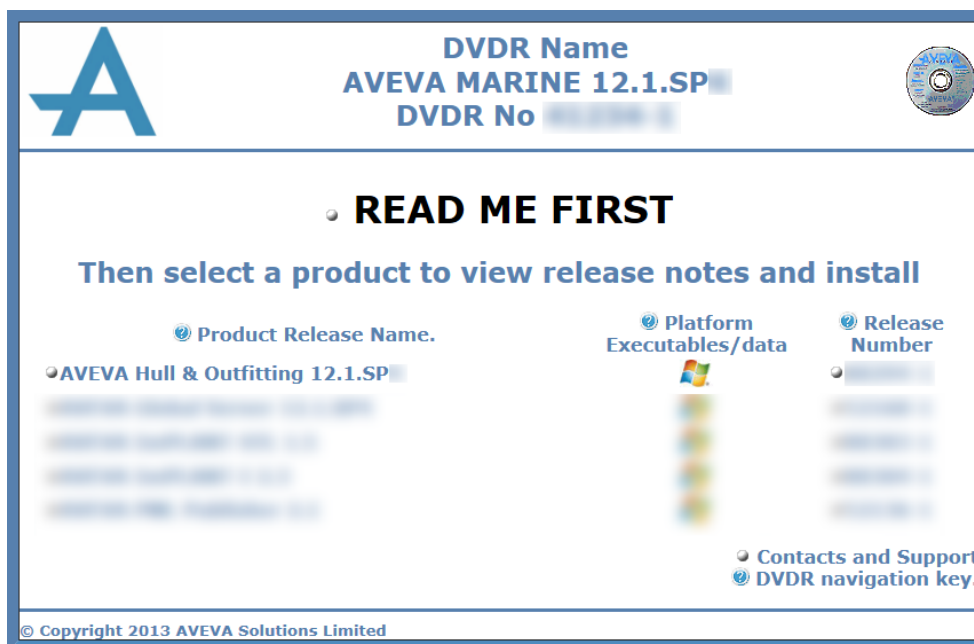
Manual installations can be easier if run from the Setup.exe program, which ships with each .msi installation supplied on disk. Each Setup.exe checks and installs the dependencies of its primary installation (like the .NET framework), before running its primary (MSI) installation. Users should never see **missing dependency** errors if installations are started from the Setup.exe program.

Setup.exe also tells the primary installation to target the C:\ drive/volume, which is the preferred default in most manual installations. However, if a more customised installation is required then running the primary MSI directly becomes the better approach. In such cases, run time dependencies (Like the .NET framework) must be deployed beforehand.

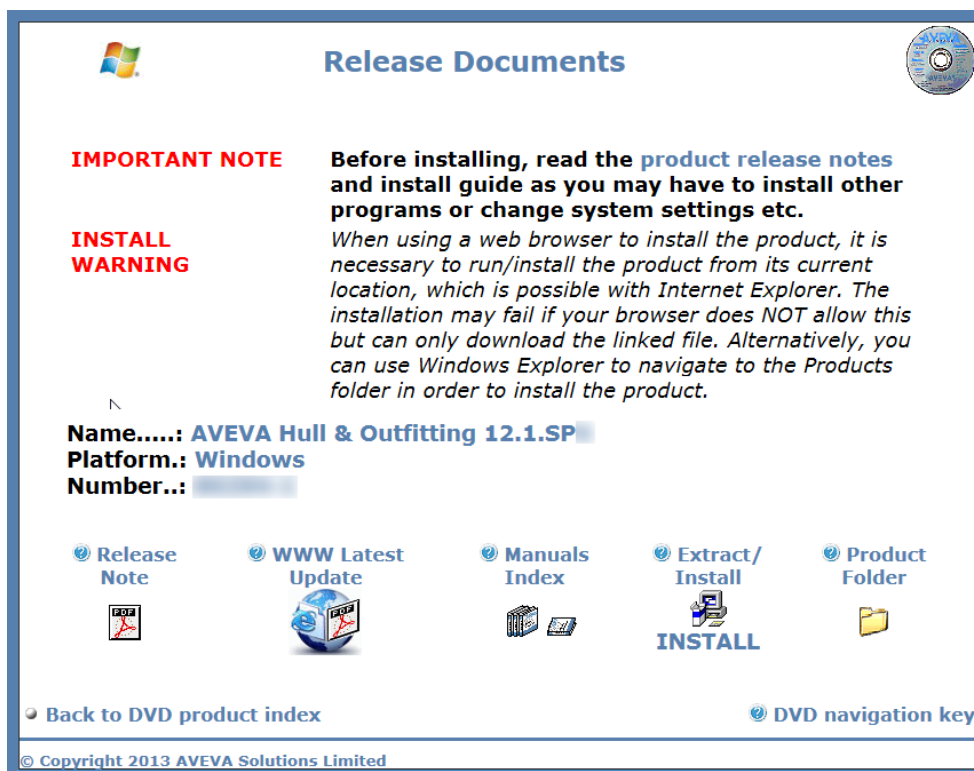
Using Setup.exe displays the **Marine Setup Wizard** window.



A screen showing the available products is displayed.



Click **AVEVA Hull & Outfitting** to display the **Release Documents** screen.



Click **INSTALL** to display the **Marine Setup Wizard**.



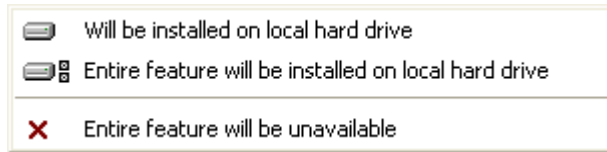
Click **Next** to display the **End-User License Agreement** window.



The user must read the End-User License Agreement and then select the **I accept the terms in the License Agreement** check box to activate **Next**.

Note: If the user does not select the check box, **Next** will not activate and the installation cannot continue.

After clicking **Next** the **Custom Setup** screen is displayed.

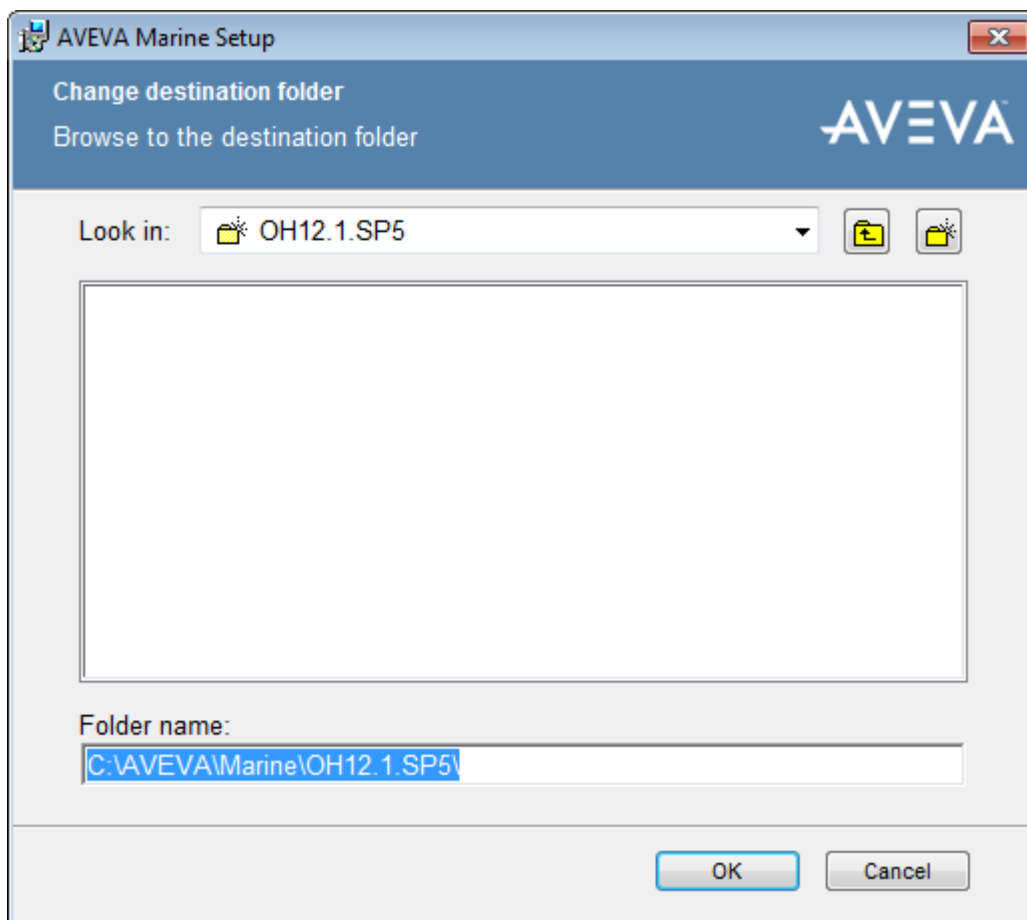


Will be installed on local hard drive	Selecting this option will install the feature locally.
Entire feature will be installed on local hard drive	Selecting this option will install the feature and all its sub-features locally.
Entire feature will be unavailable	Selecting this option prevents the current feature and all its sub-features from installing.

Clicking **Reset** will de-select all currently selected features.

If the Feature is not Unavailable (), a Location Path showing where the program is to be installed may be displayed, and if so a **Browse** button becomes active.

If a Location Path is not displayed and the Browse button is inactive when a specific Feature is highlighted and scheduled to be installed, then that Feature contains something which embeds itself inside another program (whose path is separately configurable). Clicking **Browse**, when active, displays the **Change Destination Folder** window which allows the user to select a different folder name and location path.



Addins and Addons are examples of this sort of thing, as they have no folders of their own, and they reside inside other programs.

It may appear confusing that some Features do/do not have configurable folder paths. The following list is hoped to clarify this:

Feature	Configurable	Default Path
Hull and Outfitting	Yes	C:\AVEVA\Marine\OH12.1.SP5\
Marine Sample Projects	Yes	C:\AVEVA\Marine\Project12.1.SP5\
Global Client	No	C:\AVEVA\Marine\OH12.1.SP5\
Area Based ADP	No	C:\AVEVA\Marine\OH12.1.SP5\
Multi-Discipline Supports	No	C:\AVEVA\Marine\OH12.1.SP5\
Schematic 3D Integrator	No	C:\AVEVA\Marine\OH12.1.SP5\
ExPlant-A	No	C:\AVEVA\Marine\OH12.1.SP5\
ExPlant-I	No	C:\AVEVA\Marine\OH12.1.SP5\
Initial Design	No	C:\AVEVA\Marine\OH12.1.SP5\

Marine Suite Help	No	C:\AVEVA\Marine\OH12.1.SP5\Documentation\
-------------------	----	---

Changing the Folder Path of the Feature "AVEVA Marine Suite" will affect all applications residing in subfolders. For example:

if "AVEVA Marine Suite" = J:\Test\

Then

"Hull and Outfitting" = J:\Test\OH12.1.SP5\

"Marine Sample Projects" = J:\Test\Project12.1.SP5\

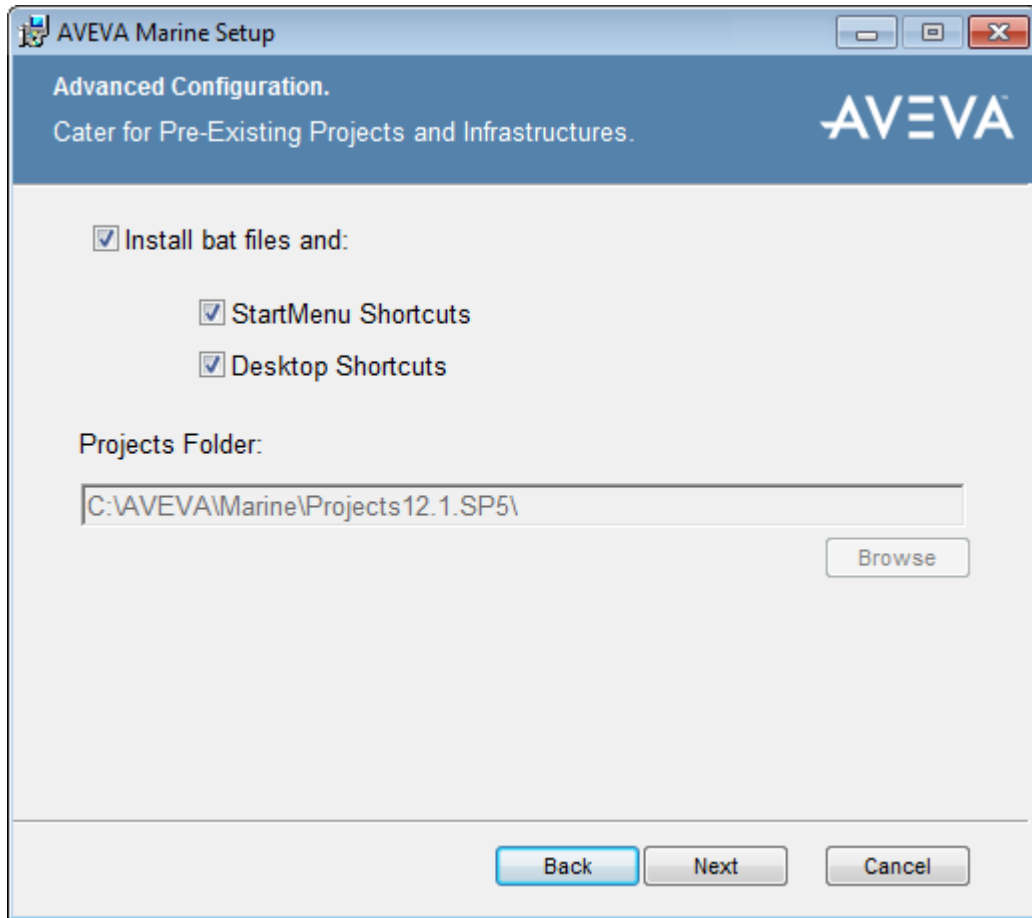
...

"Manuals" = J:\Test\OH12.1.SP5\Documentation\

However, if any of the other folders were to be directly changed, they would remain in that state regardless of changes to the "AVEVA Marine Suite" folder or other underlying folders.

To check if enough disk space is available click **Disk Usage** to display the **Disk Space Requirement** screen. Highlighted volumes indicates that there is not enough disk space available for the selected features. The user can remove some of the files from the highlighted volumes, install fewer features or select a different destination volume. Click **OK** to return to the **Custom Setup** form.

Select **Back** to step back a stage, **Cancel** to terminate the Setup Wizard or **Next** to display the existing projects and infrastructures **Configuration** screen and continue the installation.



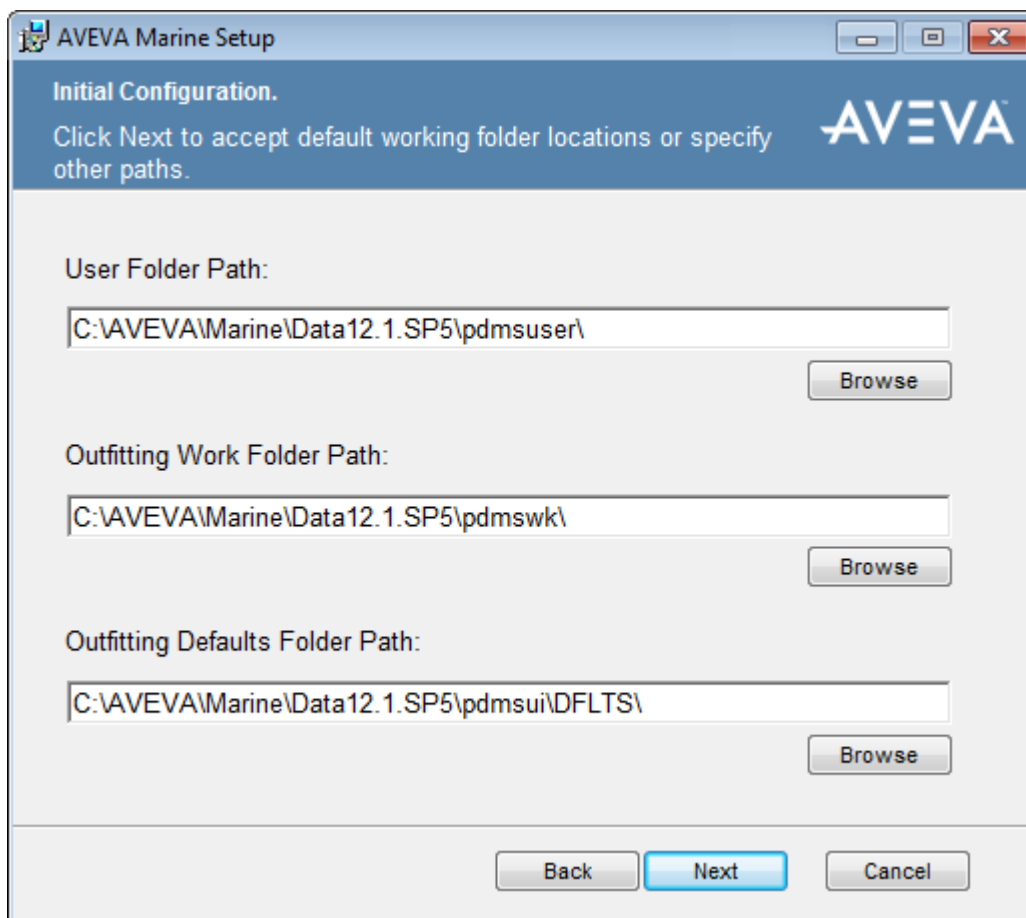
The **Configuration** screen allows the user with existing projects and infrastructures to customise the installation.

The **Projects Folder** path can only be changed if the user has chosen to install the Marine or the Marine Compatible MDS models. To change the path click **Browse** to display the **Change Destination Folder** window which allows the user to select a different folder name and location path.

Unchecking the **Install bat files and:** checkbox deactivates the **StartMenu Shortcuts** and the **Desktop Shortcuts** options. Both options are dependant on the bat files being installed.

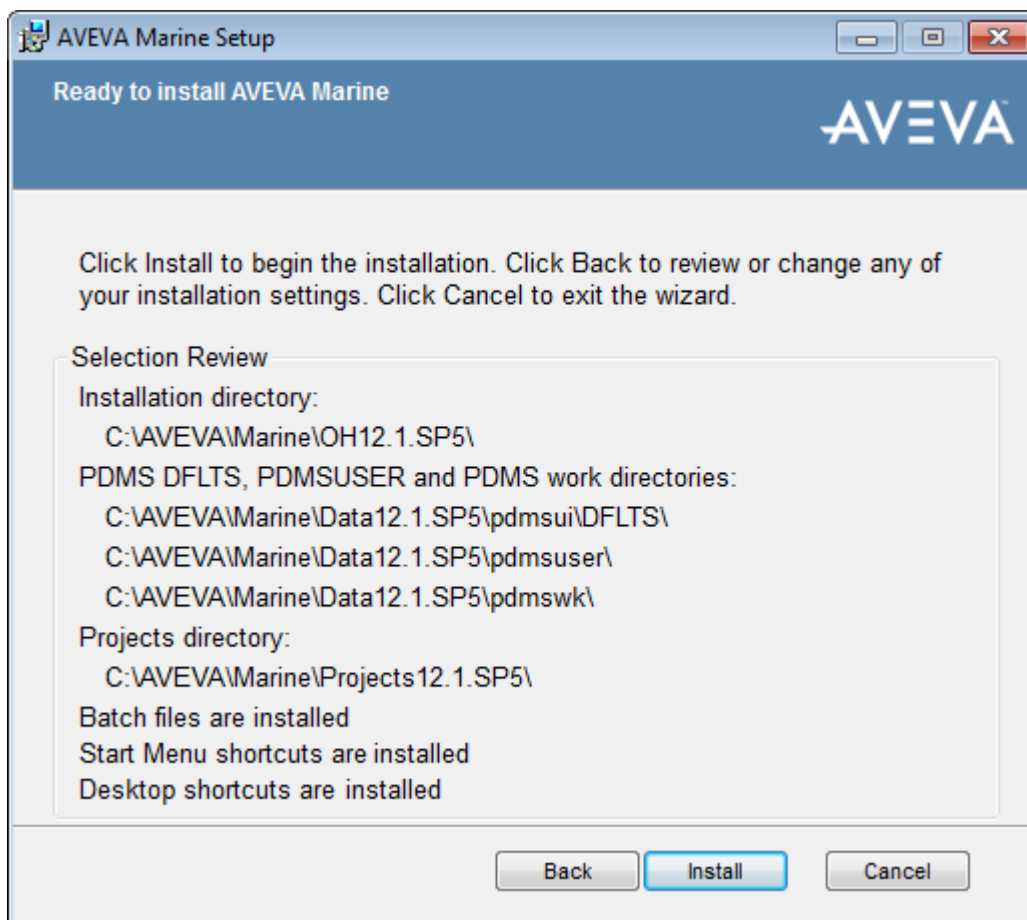
Unchecking the **StartMenu Shortcuts** checkbox deactivates the **Desktop Shortcuts** option. The option is dependant on the StartMenu Shortcuts being installed.

Click **Back** to step back a stage, **Cancel** to terminate the Setup Wizard or **Next** to display the folder path Configuration screen.

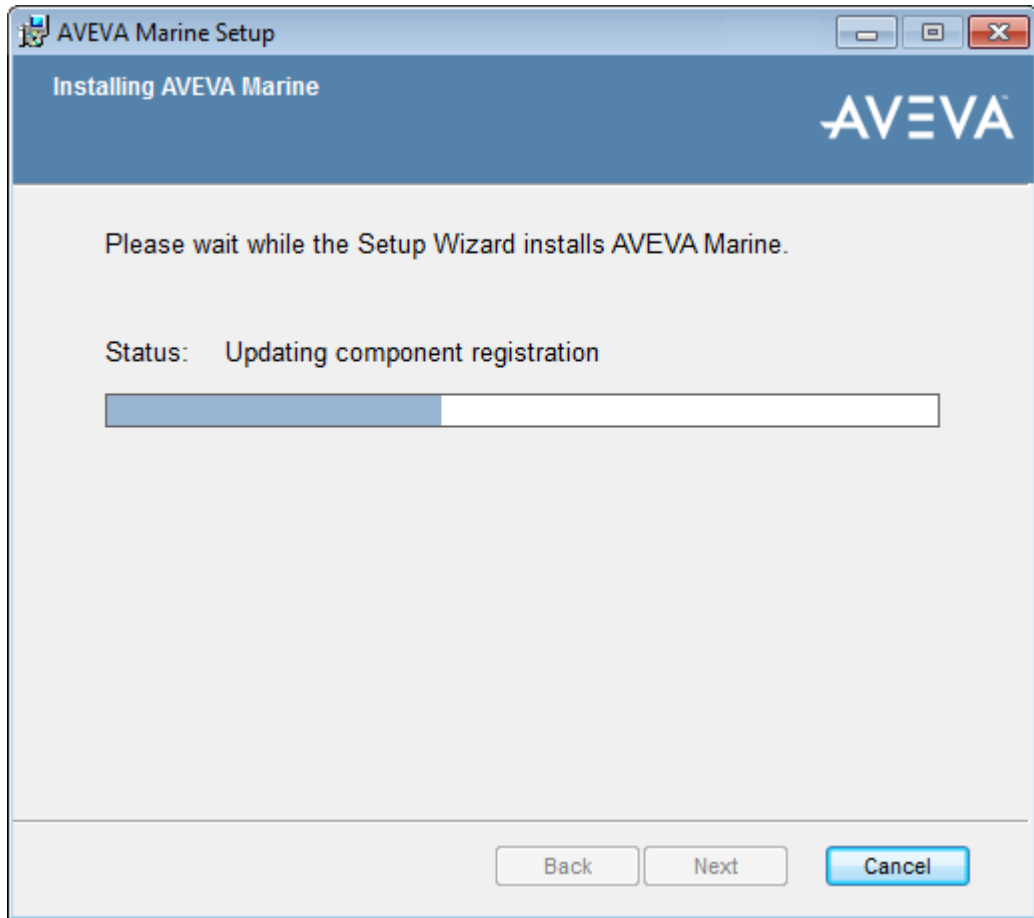


Clicking **Browse** displays the **Change Destination Folder** window which allows the user to select a different folder name and location path.

Click **Back** to step back a stage, **Cancel** to terminate the Setup Wizard or **Next** to display the **Ready to Install AVEVA Marine** screen.

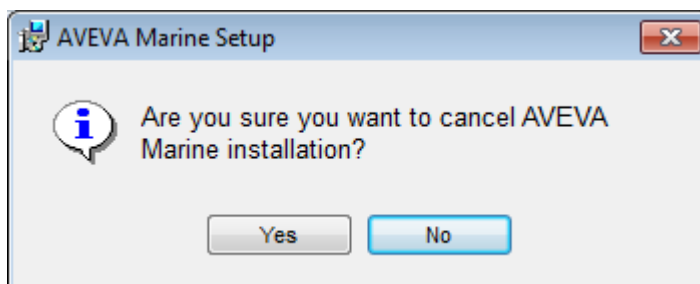


The **Selection Review** pane displays the installations options selected by the user. The user can now click **Back** to step back a stage and change their installation option, **Cancel** to terminate the Setup Wizard or **Install** to display the **Installing AVEVA Marine** screen.



The **Installing AVEVA Marine** screen displays a taskbar showing the status of the installation.

During the installation process the screen displays **Cancel**, which can be clicked to stop the installation. When selected a window is displayed asking for confirmation that the AVEVA Marine installation is to be cancelled.



Selecting **Yes** displays a screen showing the AVEVA Marine Setup Wizard was interrupted.



Click **Finish** to exit the screen and installation.

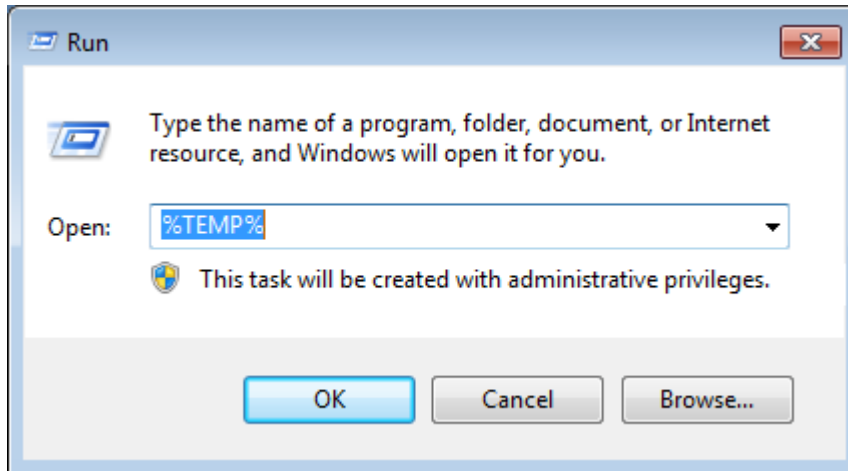
Selecting **No** on the confirm screen returns the user back to the **Installing AVEVA Marine** screen and continues the installation.



When the installation has been completed, click **Finish** to exit the Setup Wizard.

3.1 Log File

During the installation, setup.exe creates a log file of the installation in the users TEMP directory named **Marine12.1.5.log**. To locate the log file, select **Start > Accessories > Run** and type **%TEMP%** in the text field.



Click **OK** to display the users TEMP directory.



3.2 Uninstall

Complete removal of an installation can be accomplished by several methods.

- Removing it from **Start > Settings > Control Panel > Add/Remove Programs** applet
- Right clicking on the MSI file that installed it, and selecting Uninstall
- Running the command `MSIEXEC /X [Path to Original MSI]`
- Running the command `MSIEXEC /X {ProductCode of MSI}`

Removal of individual Features is also possible from the command Line, using the REMOVE property or by changing the installation state via the **Add/Remove Programs** applet.

Note: Uninstalling the Marine and associated Products installation will not uninstall Microsoft's .NET Framework or downgrade the Windows Installer Service. The .NET Framework must be removed separately if required.

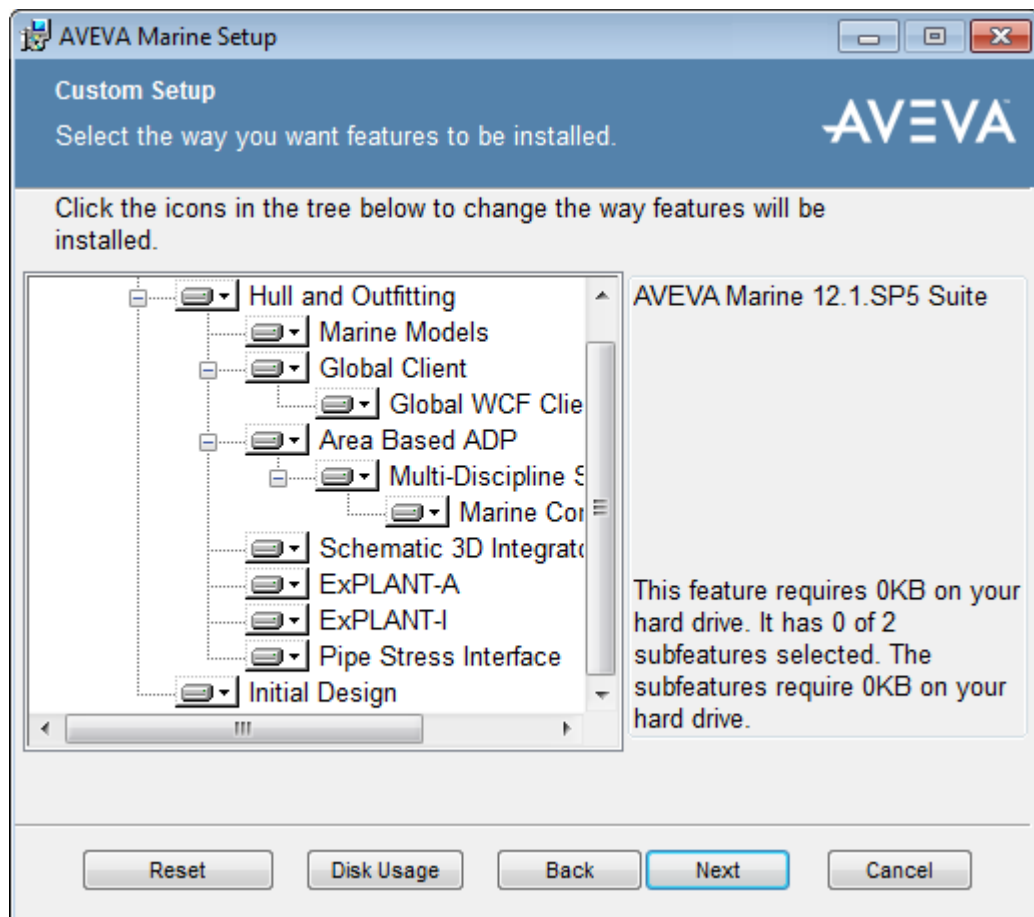
3.3 Modifying

There are several ways to modify an installed installation. For example:

- Changing it from **Start > Settings > Control Panel > Add/Remove Programs** applet
- Right clicking on the MSI file that installed it, and selecting Install, then selecting Change at the subsequent window.

- Running the command `MSIEXEC /I [Path to Original MSI]`, then selecting Change at the subsequent window.
- Running the command `MSIEXEC /I {ProductCode of MSI}`, then selecting Change at the subsequent window.

Once the **Change** option is selected the **Custom Setup** window is displayed allowing the user to change which features are installed.



The **Change** option can not be used to change the Location Path. If the user wants to change the Location Path, the current installation must be removed and then re-installed using the new Location Path.

Note: Never move files by other means, as this may trigger healing, however Copying files to other locations will not cause problems.

3.4 Repairing

If any programs stop working, or the installation has knowingly been damaged, then **Repair** may fix the problem.

There are several ways to repair an installation. For example:

- Changing it from **Start > Settings > Control Panel > Add/Remove Programs** applet
- Right clicking on the MSI file that installed it, and selecting Install, then selecting Repair at the subsequent window.

- Running the command `MSIEXEC /F [Path to Original MSI]`, then selecting Repair at the subsequent window.
- Running the command `MSIEXEC /F {ProductCode of MSI}`, then selecting Repair at the subsequent window.

Repairing installations causes them to heal themselves. For more information refer to Healing.

3.5 Healing

MSI technology has inbuilt self repairing mechanisms. As such it is generally unwise to alter file and folder names, shortcuts or registry manually, as this may trigger the MSI which deployed the files to redeploy them.

However, it is expected that some users will wish to alter Appware, "Sample Data" Shortcuts and .bat file variables. As such, AVEVA installations have been designed to minimise the ability of the MSI to heal itself in such cases.

Sample Data, Models, example projects and shortcuts, should not trigger healing if they are deleted or altered. The consequence of making shortcuts editable/deletable is that MSI Advertising will not function.

3.6 Modified New and Removed Files

Irrespective of whether an MSI is installing, changing state, or repairing, files are removed/deployed/overwritten based upon certain [file version rules](#). As such, the changes caused by a repair or an installation depend upon the initial state of the computer.

3.6.1 User Modified Appware

In MSI/Deployment terms, Modified appware files constitute un-versioned, language neutral files, whose "Created Date" and "Modified Date" differ. However, if appware files have been moved or handled in certain ways, then it is possible that the "Created Date" and "Modified Date" will **not** differ, in which case the [file replacement](#) behaviour of the installation will be different.

In the case where an appware files "Created Date" and "Modified Date" is different, no MSI will ever overwrite such a file. So Modified appware will never be upgraded (by default), changed or bug fixed by an AVEVA MSI or Patch.

In the case where a modified appware files "Created Date" and "Modified Date" does **not** differ, the appware file with the newest date will prevail (by default). This scenario is most likely to happen when installing a Patch or an MSI which is not classed as a "Service Pack" or a "Full Release", since releases with more minor designations are not side-by-side deployable, as they are intended as upgrades to pre-existing releases.

Once appware has been modified, it must fall to the author/owner to maintain the changes, since there is no reliable way to reconcile code differences. In extreme cases this may mean that AVEVA installations cannot be used to deploy appware, and those clients must make their own code merges.

3.7 Models, Sample Data and Example Projects

It is in the nature of Models, Samples and Examples, that:

- The constituent parts are often interdependent in some way.
- Interdependencies will vary between releases and can be difficult and risky to migrate automatically.
- Their file footprint is apt to change drastically.
- They will not always be required.
- They may be moved, copied and shared.
- File paths and folder names may change in time.
- Parts of them may be re-used in other scenarios.
- They can be supported or migrated across many versions of an application.
- Many different programs may work with them and with data derived from them.

Hence, it is felt that Models, Samples and Examples:

- Should be regarded as separate entities in their own right.
- Have a lifecycle which is likely to differ from the applications which they service.
- May be released with installations which install applications, but should be as easy as possible to divorce from such installations.
- Are unsuited to Repair or Patching by installations.
- Are risky to de-install/remove when applications are removed.
- Should be released holistically.
- Should be available separately from application installations in special cases.
- May suit release in a simple compressed archives (.zip file or self extracting executable say).

Current AVEVA MSI installations attempt to address these requirements by completely removing an installations ability to heal its Models, Samples and Examples. If a new copy is required, the installation must be completely removed, and then reinstalled. Repairing will not restore Models, Samples or Examples.

Uninstalls, are achieved by deleting the root folders where the Models, Samples and Examples were deployed to. It is important to remove them if they are to be retained.

3.8 Changing Default File Replacement Behaviour

It is possible to alter default [file replacement behaviour](#) in circumstances where an MSI installation or Patch is launched from a command line. This is achieved with the [REINSTALLMODE](#) property or the [/f command line switch](#). Otherwise default file replacement rules will apply.

Default file replacement rules should suit most users.

It is desirable to change file replacement behaviour, in situations where the Date/Time stamps of files might not support the required replacement of files.

3.8.1 Post Deployed .bat files

The .bat files which AVEVA installations deploy and edit are regarded as necessary for the applications to work, and so could be repaired by the installation if they are removed. However, they can be edited or replaced post deployment without risk, as long as the files "Created Date" and "Modified Date" is different (this stops an MSI from overwriting the file).

3.8.2 Pre-Deployed .bat files

It is feasible to replace .bat files before deployment, in the case where deployments are staged from Admin Installations. However the details and issues concerning Admin Installations are beyond the scope of this manual.

It is also possible to add additional files to an MSI installation, but an MSI table editor of some sort would be required. This is not a trivial option, and may well require the services of a professional re-packager. However it can be accomplished without compromising an installations ability to be patched.

3.8.3 .bat file Shortcuts

All the shortcuts pointing to the .bat files are of the non-advertised or normal type. As such they can be deleted and edited, without triggering healing.

3.9 Network (Admin) Installations

[Administrative Installation Mode](#) is a type of .msi installation, which **only** causes an uncompressed copy of the original installation to be generated to a specified location ([TARGETDIR](#)). It does not install applications, it creates another installation.

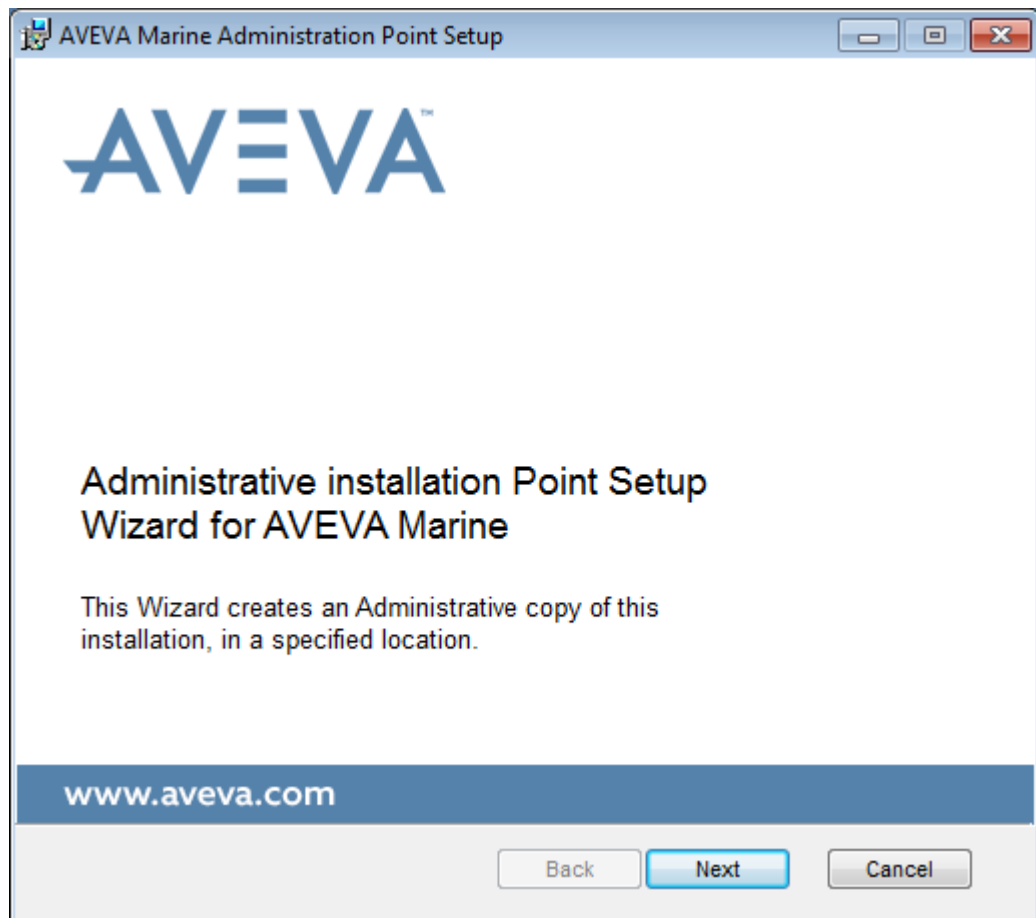
The installations media (deployable files) is unpacked into a subfolder beneath the regenerated .msi file, rather than stored in external .cab files or in .cab files embedded within the .msi file itself. The resultant folder structure created resembles that, which a (Typical) local deployment would create by default.

Administrative installations do not become installed onto the computer which spawned it. They only transform the source installation into an uncompressed variant of itself during this process.

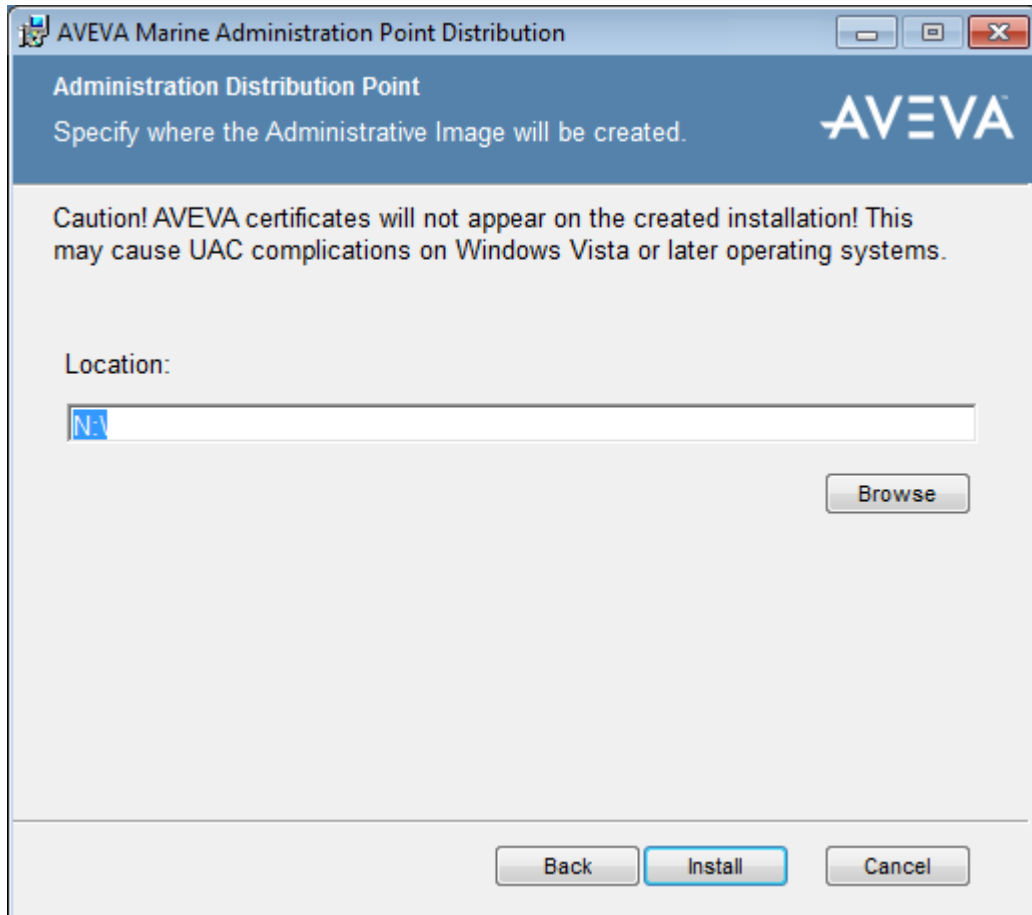
3.9.1 Create an Administrative Installation Point

To create an admin installation from an installation, start the installation from the Command Line as follows to display the **AVEVA Marine Administration Point Setup** screen:

```
MSIEXEC /A [Path to MSI File]
```



Click **Next** to display the **AVEVA Marine Administration Point Distribution** screen.



Then enter the location where the Admin Installation is to be created. Clicking **Browse** displays a window allowing the user to navigate to the required location.

Click **Install** to start the installation.

Conversely, the following Command Line will achieve the same result quietly:

`MSIEXEC /A [Path to MSI File] /QN TARGETDIR=[Path to Admin Installation]`

3.10 Issues with Administrative Installation Points

3.10.1 .Net Security

.Net security defaults to not allowing .Net program code to execute if it resides in a network location. There are several ways to "Trust" such locations, but Trusted it must be, if programs are to run across a network.

3.10.2 Signing

Newer windows operating systems have code signing embedded into their security. As such, AVEVA have begun signing .msi files and .cab files for installations which are not restricted to Windows XP or older operating systems.

Unfortunately, the Administrative Installation process recreates and/or alters the source .msi file and removes .cab files altogether, thus removing any file signatures. Changes to signed file always invalidates its signature.

In this release AVEVA have signed all dll's and exe's that are supplied. However in order to cater for installations where internet access is not permitted/available we provide an .exe.config files for each exe file that tells the operating system not to validate the digital signature. By default the installation is set to deploy the .exe.config to disable the digital signature validation. To enable digital signature validation the command line option CERTIFICATECHECK=1 should be set.

3.10.3 Using Files within the Source Installation

MSI databases can be authored to allow Features to be installed in the "Will be installed to run from network" state. AVEVA Installations do not currently support this mode; however it may be made available if requested.

The "Will be installed to run from network" option is otherwise known as run-from-network. In this mode, files are not copied onto the target computer, but registry, shortcuts and file edits do occur on the target computer, and they address the programs which reside inside the source installation.

3.10.4 Network Throughput

Admin installations can be very suitable for staging large deployments, because uncompressed installations usually travel better over networks because they are even more fragmented. The overall size is typically much larger; however this is still tends to be easier on Packet Switching Networks.

3.10.5 Advantages of a Network Installation

Once computers address files across a network, it becomes crucial that the files are accessible whenever required. This then begs questions regarding:

- Network Reliability
- Network Performance
- File Server Reliability
- File Server Performance
- Change Synchronisation
- Change Granularity

Common reasons for storing programs on file Servers are:

- To avoid the network loading of large and un-staggered rollouts.
- To guarantee the user base is working with the same files.
- To prevent tampering with programs and settings.
- To maintain a tradition.

All of these issues are of course resolvable with locally deployed applications

Note: Direct editing of .msi's is discouraged in favour of Transforms (.mst files).

3.11 Run Marine and Associated Products from Network Locations

Running programs directly from network locations is discouraged because:

- AVEVA installations have been redesigned to deploy well across networks.
- Network reliability and performance is less of a factor.
- Patch rollback functionality only works with local installations.

However, it is recognised that network based programs suit circumstances where:

- Multiple instances of the same program are required.
- An embedded tradition of working this way exists.
- Change impacts Quality Control systems.

So this information is provided to facilitate successful over network operation.

Definitive instructions on how to run AVEVA programs from network file servers are impractical, given the many ways in which AVEVA programs can interact with each other and with Project data. Not to mention other possible permutations.

So the following information is deliberately general and lacking in detail and it is aimed at assisting a highly skilled readership.

The following things must happen if programs are to be run directly from a file server:

- The programs and the projects environment must be correctly defined.
- Appropriate .NET Trust should be granted to the programs network location.
- Visual Studio 2008 C++ Runtimes should be pre-installed.
- .NETFramework Redistributable should be pre-installed.
- The appropriate shortcuts and drive mappings should be provisioned.

3.11.1 Administrative Installation Points

The Files and Folders extracted to an "Administrative Installation Point" are laid out in a fundamentally identical way to a local installation. Additionally, files which are normally deployed to operating system folders will also be visible, but will play no functional part unless deployed locally.

Some Configuration files will be incomplete, or have incorrect data in them, as they are not edited until they are deployed (Locally). These files may require manual editing and concern which Modules, Addins, Addons and "User Interface Modifications" are loaded.

3.11.2 Patch Administrative Installations

A slightly different Patch is required for an administrative installation than for a local installation, and standard AVEVA patches target local installations. If such a Patch is applied to an Administrative Installation Point, the patch will succeed. However, any local installations which have previously been deployed from it will cease to recognise it as their source installation. So as long as local deployments have not occurred, an Administrative installation is patchable with an AVEVA patch. Similarly, if programs within the installation are addressed by a clients own means, then the Administrative Installation can be patched at any time with impunity. Since in this event the Installation has not been deployed (Locally) and so only the media state has any effect.

AVEVA patches are supplied inside an executable wrapper. To apply them to an administrative image, the patch must be extracted and the appropriated command line

used. The appropriate command switch is /C which extracts the contents to the current folder.

Note: Patches cannot be rolled back from Administrative Installation Points. A copy must be made before the patch is applied, in order to rollback.

Note: Patches cause changes to the .msi file, which invalidates any security certificates applied to the installation.

3.12 Copy a Local Deployment to a File Server

Installing normally to a local computer and then copying the programs to a network location, as a way to create a network based installation has several advantages over an "Administrative Installation Point":

- The fundamental configuration file editing has happened.
- The Local installation need only include the required applications.

Note: Running a normal (Local) installation, and choosing a network drive as a target for the programs will fail to install. This is because file editing is performed with the local [System Accounts](#) credentials, which are not normally recognised by other computers.

3.13 Feature Id Reference List

Feature Id	Title	Description
MARINE	AVEVA Marine Suite	AVEVA Marine 12.0.SPx Suite
MDS	Multi-Discipline Supports	Multi-Discipline Supports (MDS) Add-on to Outfitting for the design, drawing and MTO of all kinds of supports, from a complicated framework encompassing integral hangers to a simple U-bolt.
MODELS	Marine Sample Projects	Marine sample projects for Hull and Outfitting.
PSI	Pipe Stress Interface	Software to transfer pipe-stressing data between Outfitting and CAESAR II
GLOBALUSER	Global Client	Upgrades this installation to include the Remote Procedure Call (RPC) Global Client. This is required to allow Marine Users to invoke Global commands.
ABA	Area Based ADP	Area-based Automatic Drawing Production (ABA) Add-on that automates the production of volumetric drawings of using specified templates, thus simplifying their update after a change to a model.

PIDINT	Schematic Integrator 3D	Integration application to ensure data consistency between P&IDs and the Outfitting 3D model.
EXPLANTA	ExPLANT-A	DXF export driver
EXPLANTI	ExPLANT-I	DGN export driver.
PDMSOUT	AVEVA Marine Outfitting	Product family including AVEVA Cable Design, AVEVA Room Design, AVEVA Assembly Planning, and AVEVA Space Design
HULLOUT	Hull and Outfitting	Marine Hull and Outfitting
INTDES	Initial Design	Initial Design
MARINEDOCS	Marine Documentation	Marine Documentation
MDSMOD	Marine Compatible Models	Sample data, including catalogues, for Outfitting and associated products.

3.13.1 Feature Tree Hierarchy

The following diagram shows the Marine installations SelectionTree by Feature Id's. Ancestor (Parent) Features to the Left, Descendant (Child) Features to the Right.

```

HULLOUT -- MODELS
HULLOUT -- GLOBALUSER
HULLOUT -- GLOBALUSER -- GLOBALWCF
HULLOUT -- PIDINT
HULLOUT -- ABA
HULLOUT -- ABA -- MDS
HULLOUT -- ABA -- MDS -- MDSMOD
HULLOUT -- EXPLANTA
HULLOUT -- EXPLANTI
INTDES
  
```

3.14 SelectionTree Selection & De-selection Rules

Successful Inclusion of a descendant Feature implies that all ancestor Features are also included. For example, suppose the MDS Feature is required. This implies that MARINE, HULLOUT and ABA must also be included, since they are ancestors of the MDS Feature (see above).

De-selection of a Feature overrides any commands or defaults which may have otherwise included it. For example, suppose MDS is included, and ABA is excluded. The result will be that MARINE and HULLOUT are installed, since they default to being installed. Removing ABA causes MDS to be removed, since ABA is its Parent (Ancestor), (see SelectionTree Hierarchy diagram).

3.15 Selecting Features from the Command Line

Features can be in the following states:

- Run Locally
- Run from Source
- Not Present
- Advertised

There are several ways of controlling Feature states, but for the sake of clarity this discussion will be limited to the [ADDLOCAL](#) and [REMOVE Properties](#). The [ADDSOURCE](#) Property is considered less relevant, as the "Will be installed to run from network" Feature option, is not available to Features within this installation.

The [ADDLOCAL](#) and [REMOVE Properties](#) are comma separated lists of an installations' Feature Id's. Any Features intended to be "Run Locally", will be listed in the [ADDLOCAL](#) Property, whilst any Features intended not to be installed will be listed in the [REMOVE](#) Property.

As previously stated, de-selection commands override inclusion commands. The reason for this is because the [REMOVE](#) Property is evaluated after the [ADDLOCAL](#) property. The partial list below shows the order in which FeatureState [Properties](#) are evaluated by the MSI Service:

1. ADDLOCAL
2. REMOVE
3. ADDSOURCE
4. ADDDEFAULT
5. REINSTALL
6. ADVERTISE

3.16 Command Line Definable Directories

The following directories can be set on the Command Line, in the same way that Properties are assigned.

Directory Id	Explanation
PDMSUSER	Specifies the pdmsuser directory location
PDMSWK	Specifies the pdmswk directory location
DFLT	Specifies the PDMSUI\DFLT directory location
PROJECT	Specifies the directory to extract the models to
PRODUCTDIR	Specifies the root product directory location
TARGETDIR	Target location for an Administrative Installation
ROOTDRIVE	The Partition where All Applications will be targeted (unless application directories are set).

As with Properties, Directories for which the Id's are all uppercase, can be re-defined on a Command Line. They must also be addressed in uppercase on the Command Line, because Id's are case sensitive.

e.g. PRODUCTDIR="C:\Some Path with Spaces in it"

Will make Marine install to:

"C:\Some Path with Spaces in it"

Rather than to:

C:\AVEVA\Marine\OH12.1.SP5

Note: There is no space around the equals (=) sign.

Note: The quotes surround the value, and would be unnecessary if the value had no space within it.

3.16.1 Sample Command Lines

A most definitive reference to MSI Command Line arguments can be found [here](#). The following examples have been tested, but are only intended to demonstrate general Command Line principles.

Example 1 Installing Everything with Exceptions

Installs all Features except "Initial Design" and "Area Based ADP" to the "E:\\" drive

Unattended Installation showing a progress bar with no cancel button, no finished dialog and standard logging. Never attempt to Reboot

Create a basic log

```
MSIEXEC /I [Path to MSI] ADDLOCAL=ALL REMOVE=INTDES, ABA
ROOTDRIVE=E:\ REBOOT=ReallySuppress /l [Path to log file] /qb!-
```

Example 2 Installing projects to a different location and setting users writeable areas to a separate drive

Installs all Features to the C:\

```
MSIEXEC /I [Path to MSI] ADDLOCAL=ALL ROOTDRIVE=C:\
PDMSUSER=D:\AVEVA\Data12.1.SP5\PDMSUSER
PDMSWK=D:\AVEVA\Data12.1.SP5\PDMSWK
DFLTS=D:\AVEVA\Data12.1.SP5\PDMSUI\DFLTS
PROJECT=Q:\PDMS\12.1.SP5\l [Path to log file] /qb!-
```

Example 3 Removing an Installation

```
MSIEXEC /X [Path to MSI] /QB!
```

Example 4 Removing a Feature

Remove "ExPLANT-I"

```
MSIEXEC /I [Path to MSI] REMOVE=EXPLANTI /qb!-
```

3.17 AVEVA Database Cache Service

Before the Cache Service can be used it is necessary to activate Microsoft Message Queuing (MSMQ) on your machine.

3.17.1 Activating Message Queuing (MSMQ)

The manual installation is achieved using the **Start > Control Panel > Add or Remove Programs** facility. The steps required to do this are shown in: [Figure 3.1.: Steps to activate Message Queuing on Windows XP](#) for Windows XP; [Figure 3.2.: Steps to activate Message Queuing on Windows 7](#) for Windows 7.

Note: You must have Administrator rights to be able to activate MSMQ.

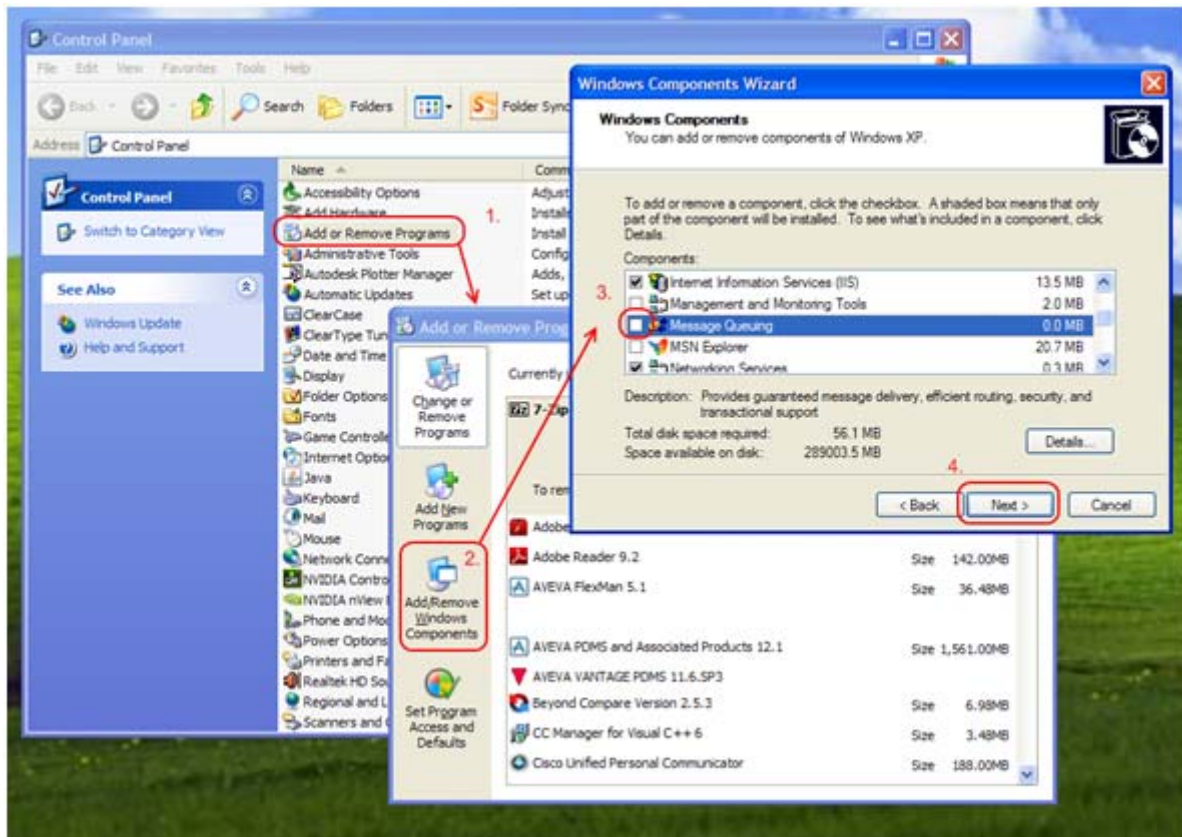


Figure 3.1. Steps to activate Message Queuing on Windows XP

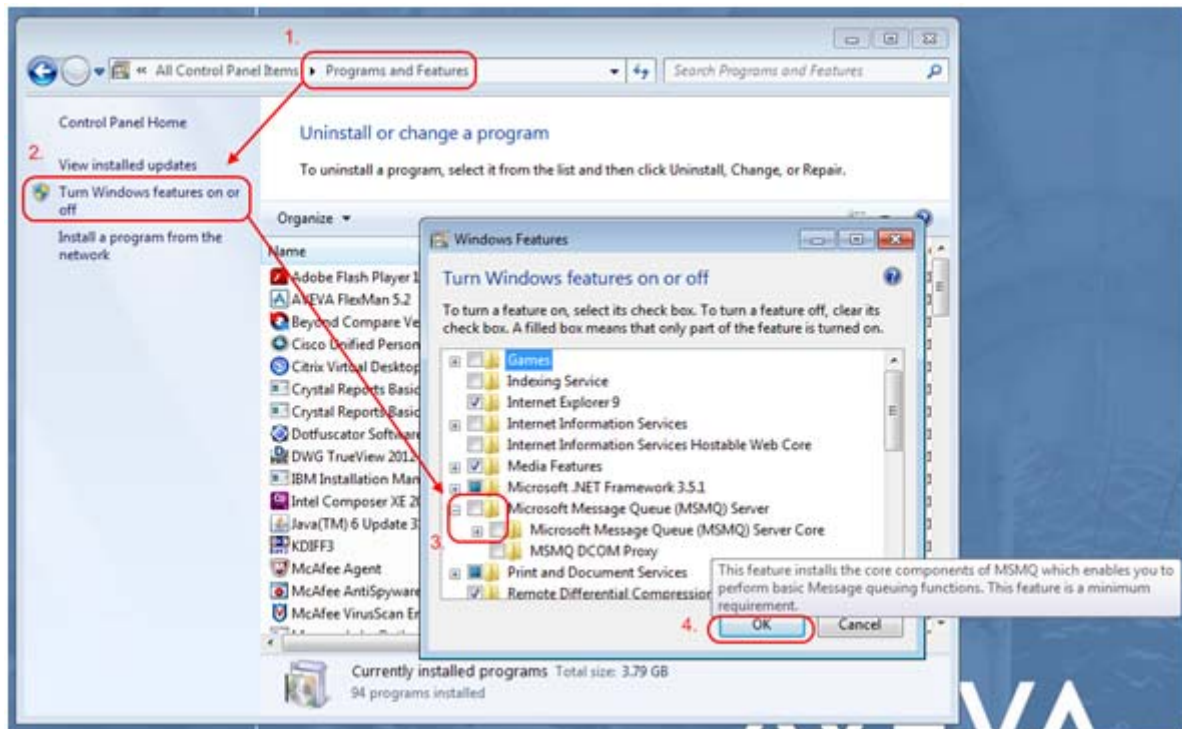


Figure 3.2. Steps to activate Message Queuing on Windows 7

3.17.2 Installing the Cache Service

The Cache Service should be installed using the installation wizard. During the installation steps it is possible to specify:

- The folder location in the available NTFS file system in which to store the service and management tool and related libraries.
- The folder location in the available NTFS file system in which to store project data cache files. This folder should be on a local disk with some free space. Overall application performance will benefit from fast access speeds to this folder. This folder location can be changed later using the Cache Service management tool.
- The maximum disk usage for the disk holding the cache storage folder; the Cache Service will continue to store cache data only if the disk total usage remains below this threshold. This threshold can be changed later using the Cache Service management tool.
- The number of days after which the Cache Service will automatically remove cached project data that has not been accessed in that period. This period can be changed later using the Cache Service management tool.
- Whether the Cache Service should be active immediately after installation. The Cache Service can be repeatedly activated or de-activated later using the Cache Service management tool.

4 Set Up the Windows Environment

Setting up the Windows Environment section describes how to configure the Windows operating system before running AVEVA Marine. Following installation of AVEVA Marine, the user should perform the following steps:

- Check the settings for the graphics board: See [Graphics Card Settings](#).
- Make sure the Aveva Licensing System is running. Refer to [Set the License File Location](#).
- Configure Windows parameters for Marine: Refer to [Marine Environment Variables](#).
- The user can then run Marine by clicking on the Marine shortcut or icon, which will set up the remaining environment variables required.

Note: In the following instructions it is assumed that AVEVA Marine has been installed in folder **C:\AVEVA**. If it has been installed elsewhere, substitute **C:\AVEVA** with the name of the installation folder.

4.1 Graphics Card Settings

The user should set the display to a resolution of at least 1024 x 768 (1920 x 1200 recommended) and the colour depth to True colour or 16.7 million colours.

To set the display settings, open the **Control Panel**. Select **Display** and select the **Settings** tab. Make the appropriate selections on the form for the graphics card.

Due to the rapidly changing list of hardware accelerated OpenGL graphics cards commercially available, an up-to-date list of supported graphics cards, graphics device drivers and recommended device driver configuration cannot be included in this document. However, the AVEVA Support Web pages <http://www.aveva.com/support> contain a current list of both tested and supported graphics cards, along with advice on configuration of graphics device drivers.

4.2 Set the License File Location

Before the user can use Marine, the AVEVA Licensing System must be set up. Refer to the AVEVA Licensing System guide for further information. A license file should be requested from AVEVA Customer Services.

4.3 Marine Environment Variables

AVEVA Marine uses Windows Environment variables to locate folders containing Marine system files and users project data. This allows you to locate these folders anywhere on the

file system. The location for Marine system files is C:\AVEVA\Marine\OH12.0, unless the default location was changed during the installation.

Marine is supplied with a file **evars.bat** in the Marine executables folder which sets default values for the environment variables so that they point to the folder structure as installed. It also sets the **PATH** variable to include the PDMSEXE folder containing Marine executables.

You may need to modify this file to reflect the folder structure of your own installation. The following is a list of the variables you may need to reset, with a brief description of each one and the default setting.

Note: That all environment variables must be UPPERCASE.

Search Paths of Folders Containing Spaces

Search paths of folders can contain spaces, by using a semicolon as the separator between folders, for example:

```
set PDMSUI=C:\Local Marine\AppData;M:\Project
Marine\AppData; a:\Company Marine\AppData
```

Note: That in order to define an environment variable that contains just one item including spaces, it must explicitly include a final semicolon. For example:

```
set PDMSUI=C:\Local Marine\AppData;
```

Length of Search Path Environment Variable

The lengths of Marine search path environment variables can be up to a maximum of 5120 characters.

Recursive Expansion of Environment Variables

An environment variable defining a search path can itself contain references to other environment variables. For example:

```
set MPATH=%ROOT%\Standard Version;%ROOT%\Update Version
set ROOT=C:\Program Files\AVEVA; (note final semicolon)
```

Then on PDMS:

```
*$m%MYPATH%\mac
```

Will check in turn for:

```
C:\Program Files\AVEVA\Standard Version\mac
C:\Program Files\AVEVA\Update Version\mac
```

System Environment Variables

The following environment variables refer to the location of Marine system files.

PDMSEXE	Executable folder C:\AVEVA\Marine\OH12.x.xx\
PDMSEXEMAIN	Copy of MarineEXE, but can be a search path C:\AVEVA\Marine\OH12.x.xx\

PDMSUI	Application User Interface folder (PML 1) C:\AVEVA\Marine\OH12.1.SP5\PDMSUI
PMLLIB	Application User Interface folder (PML 2) C:\AVEVA\Marine\OH12.1.SP5\PMLLIB
PDMSDFLTS	Applications defaults folder C:\AVEVA\Marine\OH12.1.SP5\PDMSUI\dfits
PDMSPLOTS	Applications PLOTS folder C:\AVEVA\Marine\OH12.1.SP5\PDMSUI\plots
PLOTADC	Pointer to the PLOT executable. C:\AVEVA\Marine\OH12.1.SP5\plot\plot.exe
PDMSHELPPDIR	Online help data files. C:\AVEVA\Marine\OH12.1.SP5\Documentation

User Environment Variables

The following environment variables refer to the location of users Marine folders. Users must have write permission to these folders.

PDMSWK	Temporary Marine work file folder C:\AVEVA\Data12.1.SP5\PDMSWK
PDMSUSER	Application Users save/restore setup area (<i>or the users 'home' folder</i>) C:\AVEVA\Data12.1.SP5\PDMSUSER\
PDMSREPPDIR	Marine Applicationware Report Input and Output folder. C:\AVEVA\Marine\OH12.1.SP5\PDMSUI\reports
PDMSDFLTS	Applications defaults folder. Note that individual users may not have permission to change Marine System defaults in C:\AVEVA\Data12.1.SP5\PDMSUI\dfits, but they should have their own local defaults area. This is achieved by setting a MarineDFLTS search path, e.g. 'C:\USERDFLTS C:\AVEVA\Data12.1.SP5\PDMSUI\dfits'.

Project Environment Variables

For each Marine project, the following environment variables must be set. For example, for a project ABC:

ABC000	Project database folder
ABCISO	Project ISODRAFT options folder
ABCMAC	Project Inter-DB connection macro folder
ABCPIC	Project DRAFT picture file folder
ABCDRG	Hull drawings
ABCDWG	Outfitting drawings

ABCMAR	Project Hull database folder
ABCDIA	Diagrams
ABCSTE	Stencils
ABCTPL	Templates
ABCDFLTS	Project configuration settings

Marine is installed with sample project data. Marine Project Environment variables for these projects are set in evars.bat.

4.4 Run Marine from a Command Window

The user can run Marine from a Command window rather than using the Start/Programs menu. To do this the user must set your **PATH** variable to include the folder where Marine is installed.

Windows XP

- Open the **Control Panel**.
- Select **System**, and pick the **Advanced** tab.
- Click on the **Environment Variables**.
- Edit the system variable **Path** to add the Marine folder (for example, C:\AVEVA\Marine\OH12.1.SP5) to the list of folders in the Path variable.

Once the Path variable includes the Marine folder, you can run Marine by typing marine at a command prompt.

5 Run Marine Under Windows

5.1 Spaces in Windows Pathnames

To run macros with spaces in the filenames, the user must enclose the filename in quotes, without a leading slash. For example:

```
$m'c:\Program Files\Macros\space.mac' arg1 arg2
```

Due to the way in which Marine parses its command lines this format is also recommended for use with filenames containing Asian characters.

5.2 Link to Microsoft Excel

The user can make your reports load directly into a Microsoft Excel spreadsheet, as comma-separated files:

- Select **Utilities > Standard Reporting > Reports > Modify** from the Marine main menu bar.
- Pick a report template.
- Click **Open** to display the **Modify Report Template** window.
- Select **Options > Other Formats** from the **Modify Report Template** window menu bar to displays the Report Format window. From the Report Format window:
 - Choose the **CSV** option from the **Format** drop-down list.
 - Click **OK** to return to the **Modify Report Template** window.
- Select the **Overwrite** radio button.
- For the filename enter %PDMSUSER%\fname.csv (the .csv extension is important).
- In the System command text box enter.
- 'cmd/c "C:\<excel folder>\excel.exe" %PDMSUSER%\fname.csv'.
where C:\<excel folder> is the location of Excel on your system.

Note: It is necessary to enclose Windows command arguments in double quotation marks if the argument contains embedded spaces. :<excel folder>.

5.3 Windows Parameters

The following setting can be used to improve performance when repositioning forms in Marine. These are optional Windows Parameter settings.

Windows XP

- Open the **Control Panel**. Select **Display**, then select the **Appearance** tab, click **Effects**. On the **Effects** window, uncheck **Show Window Contents While Dragging** and **Show shadows under menus**.

6 Configure Area Based ADP

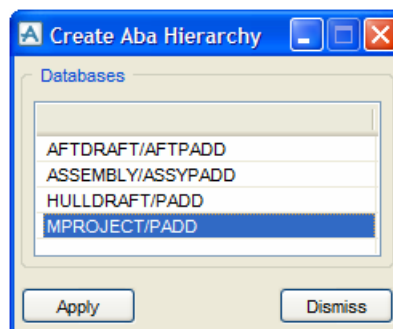
Before starting the configuration process make sure the project has been set up with a Drafting database (PADD) for the administrative elements of ABA and at least one Drafting database (PADD) for the user elements. Make sure that access to the Drafting administration database is controlled via the Drafting administration team.

1. Enter the Drafting module using the MDB for ABA administration and be sure to enter via Macro files, NOT Binaries.
2. Once in the Drafting module select the **Draft > Administration** from the Draft menu and then **Draft > Area Based ADP**.
3. Select the **Settings > Area Based ADP Defaults** from the menu.
4. When the defaults form appears it shows the settings in the supplied DRA-ABA-DEFAULTS file. This is the hierarchy required for running the ABA application. For a detailed explanation of the elements defined in the Defaults file, refer to the [Draft Administrator Application User Guide](#).

Hierarchy Positioning			
Library	/Aba/graphical_reports_library	CE	LIBY
Template	/Aba/drawing_template_library	CE	LIBY,DRTMLB
Area	/Aba/area_definition_library	CE	LIBY
Classification	/Aba/disciplines_library	CE	DLLB
Keyplans	/Aba/keyplans_library	CE	SHL
Symbols	/Aba/symbols_library	CE	SYLB
Labelling	/Aba/tag_rule_library	CE	TRLB
Tasks	/Aba/task_library	CE	TASKLB

The fields in the defaults file are as follows:

Defaults file	Specifies the location directory and name of the defaults file used to set or store the ABA default settings.
Library	The top-level LIBY element which holds all the required ABA administrative elements.
Templates	Templates are stored below a DRTMLB, which is a library used to hold ABA drawing templates (DRTMPL). If more than one Drawing template library is required, this should be set to a LIBY element and all DRTMPL placed below it.
Areas	A LIBY for DLLBs which are used as drawlist libraries which hold all the IDLI (ID List) elements that store the physical limit co-ordinates for each of the ABA geographical areas.
Classifications	a DLLB used to contain all the IDLI elements for each ABA classification. A Classification is a set of database elements that have something in common e.g. Equipment elements with a PURPOSE of PUMP.
Keyplans	A SHLB is a sheet library used to hold all the OVER (overlay) elements for the keyplans.
Symbols	A SYLB is a symbol library that contains all ABA symbols e.g. north arrows.
Labelling	A TRLB is a tag-rule library that contains all the ABA tag rules.
Tasks	A TASKLB is a task library, which holds all the ABA tasks e.g. plot - send drawing to a plot file.
Create Hierarchy	This option is used to automatically create the administrative hierarchy specified in the Defaults form.



To use this functionality:

Modify the names to the desired settings.

If the required structure exists, the Create Hierarchy button will show 😊

However, if it does not exist, the button will show 😞.

Note: This only detects the existence of the LIBY, but not the elements contained within it

To create the hierarchy specified in the defaults form, click the 😞 button and select a db from the **Create Aba Hierarchy** form that appears.

The previously mentioned process will also create some initial SYMBol elements. The button icon will then turn to 😊, which indicates that the structure is OK.

Logfile Location

Gives the directory path where the log files for drawings created using ABA are to be stored.

Batchfile Location

Gives the directory path where the batchfiles created by ABA are to be stored.

The default settings for the Batch User can be set by clicking **Batch User** to display the **Batch Username and Password** form.

User Name

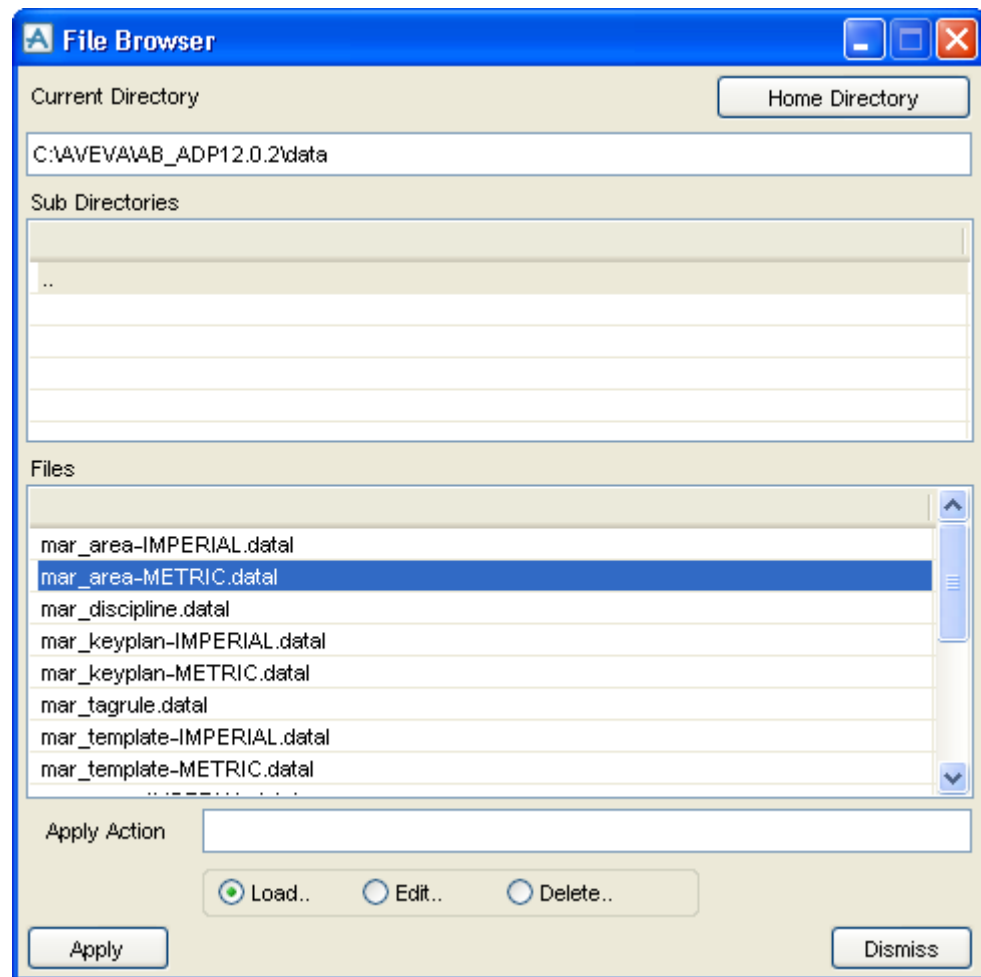
Specifies name of user that will enter the Base Product.

Password

Specifies the password of the user.

The application is supplied with a series of example data files that can be used to make a test run of ABA using the standard supplied AVEVA sample project SAM.

- The administrator also has the facility to easily load these supplied example data files. They are accessible by clicking **Input File** which displays the File Browser form.



Any data files must be loaded in the same order as shown in the Files list, since some elements rely upon the existence of other elements. To load a file, select the one the user wants to load from the Files list and click **Apply**. Repeat this until all of the necessary library files are loaded and click **Dismiss**.

Note: It is not possible to mix Metric and Imperial libraries since they utilise the same element names. If the user wants to mix them the user will need to modify the source data files.

5. In the SAM project load each of the following files in order by selecting them in the **Files** list and then press the **Apply** button.

sam_area-METRIC.datal	builds a series of example limit areas
sam_discipline.datal	builds a series of example disciplines
sam_keyplan-METRIC	builds the keyplans used by the relevant tasks
sam_tagrule.datal	builds a series of example tagrules
sam_template-METRIC.datal	builds a series of sample templates

6. For an Imperial project, load the substitute files marked IMPERIAL

Once these data files have been loaded, you can run ABA and produce some drawings.

ABA is now configured. Drafting administrators can now create the required administrative elements (areas, classifications, tasks, labels, and drawing templates) for use by DRAFT users.

Refer to the *ABA Administrator* and *User Guides* for further information.

7 MDS Installed Configuration

MDS is installed directly into the main installation directory instead of being installed in a separate folder

Note: This is also the case for ABA

As a result of this change MDS is now run using the standard set of shortcuts that are installed for the main product. This means that if MDS has been installed then it will be available for a project design or outfitting session providing that the project has been set up for MDS use (as described in the administration guide) and there is a valid MDS license available to the user.

7.1 Manual Configuration of the Product and Product Environment Variables

All MDS environment variables are set in the evars.bat file in the main installation directory. These are defined initially during the MDS installation process however.

- if the installer wants to create their own environment variable files

OR,

if at the time of install, the following is true,

- any modifications are made to the product internal structure/naming
(This product has the same internal structure/naming as the base product, eg.)
(...\\product\\pmlib has same function as ...\\base\\pmlib)

then the evars need manual (re-)definition and must be modified as follows.

7.1.1 Set Environment Variables

Environment variables modifications are made to the base product's files evars.bat from pdms.bat.

Some of these environment variables can be "searchpaths".

AVEVA "searchpath" variables are modified from the initial setting %EnvVar%, using a blank as the separator and usually PREfixing the new value, in the form.

```
SET EnvVar={newEnvVarValue} %EnvVar2% %EnvVar%
```

This will force items in newEnvVarValue to be used in preference to those in %EnvVar2% or %EnvVar%

Searchpath variables need only be changed if the administrator wishes to include additional PML code paths to their installation or otherwise include alternative locations for known files.

7.1.2 Modification of Evars.bat for MDS

The following environment variables are set up for MDS in the standard product evars.bat file:

mdsplots	Folder containing the plot files of MDS standard support types.
Licadmsexex	sets the executable location for the Licad software interface if used.
PsImdsexex	sets the executable location for the PS designer software interface if used.
Cplmdsexex	sets the executable location for the Witchhanger software interface if used.

If the user modifies the pmlib configuration after the install, the first operation when first running the product is to select the Command window and run the command,

```
"pml rehash all"
```

Note: Write access is required to the PMLLIB folders to enable the several pml.indx files to be updated.

7.1.3 Environment Variables for Projects

Except for the standard sample projects supplied with the products, the install cannot create the environment variables for users projects.

When the user creates projects they must set these adding into one of,

- the user's Environment variables set (not recommended)
- as instructions in one of the startup batfiles

The standard installation of MDS installs all of the sample MDS projects into the same folder as the standard projects folder. Each of the sample projects (detailed in [Projects](#)) contains its own environment variable batch file (i.e. evarsMDS.bat) that is called from the standard evars.bat file.

7.1.4 For Projects Used with MDS

Note: MDS defines two additional environment variables, 'mdsdflts' & 'mdsinfo', for the AVEVA project folders ...MDS\mdsdflts and ...MDS\mdsinfo used by MDS.

All other projects, which will be required to use the MDS applications, will need to have the two extra folders created, and the relevant environment variables set to point to these two new directories in the relevant evars batch file, e.g.

For a project called 'abc' create the two required folders by copying the .../MDS/ mdsdflts and .../MDS/mdsinfo folders to where the 'abc' project folders are located, and rename them accordingly, i.e. rename the copied folders mdsdflts and mdsinfo to abcdflts and abcinfo respectively.

Project Variables for MDS

mdsreports	destination of the MDS report files.
Mds000	project MDS, 000 files
Mdsmac	project MDS, mac files
Mdspic	project MDS, pic files
Mdsiso	project MDS, iso files
Mdsdflts	project MDS, dflts files
Mdsinfo	project MDS, info files

and similar project environment variables for any other projects.

7.2 Projects

7.2.1 Projects Released with this Product

There are five standard projects released with the Multi Discipline Supports product, these are:

CPL	Carpenter and Paterson Catalogue Project
LIS	LISEGA Catalogue Project
MDS	Multi Discipline Supports project containing the catalogue component, catalogue application defaults and the design databases for use with the MDS application
MDU	Multi Discipline Supports project containing the catalogue database for the User defined MDS Ancillaries available in 12
PSL	Pipe Supports Ltd Catalogue Project

8 Troubleshooting

Here are some suggestions about solving some problems you may encounter when using Marine under Windows.

8.1 General

- | | |
|---------------------|--|
| Error: | The dynamic link library DFORMD.dll could not be found in the specified path. |
| Explanation: | This is supplied with Marine. Check the PATH variable has been set. The PATH setting must include 'C:\AVEVA\Marine\OH12.1.SP5' |
| Error: | STOP. The application failed to initialise properly (0xc0000135). Click on OK to terminate the application. |
| Explanation: | This occurs after a number of other errors. The most likely reason is a problem with the PATH variable. |

8.2 Entering Marine

- | | |
|---------------------|--|
| Error: | (43,9) Environment Variable not defined for project name. |
| Explanation: | The environment variable for the project has not been set, e.g. XYZ000. |
| Error: | (43,1) Project <xxx> not available!
Error 7 allocating system database. |
| Explanation: | The environment variable for the project does not point to the correct location. Either the disk and/or folder do not exist or are not readable. |
| Error: | <username> not found. Invalid username or password.

or
Bad Password |
| Explanation: | The user is trying to enter Marine without a valid Marine username and/or password. |

Error:	PDMSWK environment variable not set.
Solution:	Set PDMSWK to c:\temp or some other folder where Marine can create temporary files.
Error:	Error 34 opening variables file.
Explanation:	The user cannot create a file in the location pointed to by environment variable PDMSWK.
Error:	Project <xxx> is pre version 8.
Explanation:	This can happen if the user has transferred the project from UNIX using FTP in ASCII rather than Binary format.

8.3 Problems with Graphics

Error:	Cannot pick certain items in DESIGN 3D shaded views.
Solution:	There is probably a problem with the display driver for the graphics board that the user is using. Make sure the user has the version of the display driver recommended on our supported display drivers Web page.
Error:	Incorrect colours.
Solution:	Check the setting of the users Display. The colour resolution should be set to 'True Colour' or 16.7 million colours.
Error:	Graphics speed (for example in 3D views) is poor.
Solution:	Check the setting of the users Display.

8.4 Troubleshooting the .NET Installation

8.4.1 Version Error

Error:	System.IO.FileLoadException: The located assembly's manifest definition with name 'MarineResources' does not match the assembly reference.
Solution:	All .Net assemblies must be the same version. Check File>Properties>Version Assembly Version on all assemblies is 12.0.0.0 or whatever the current version is.

8.4.2 Wrappers or a Dependency Missing

Error: An exception 'System.IO.FileNotFoundException' has occurred in mon.exe.

Additional information: File or assembly name Wrappers, or one of its dependencies, was not found.

Solution: All the .NET assemblies are Private assemblies and are installed under %PDMSEXEXE% (Assembly Base directory) or a subdirectory below. .NET loads private assemblies from the base directory of the application. If it can't find the private assembly here, it uses the application's configuration file called <module>.exe.config to see whether it contains any path hints. Path hints are stored in the <probing> element. So, for example, marhdes.exe.config has the following probing path <probing privatePath="Design" />. This means that Wrappers.dll is loaded from the subdirectory DESIGN below the application directory. Each Wrappers.dll depends on the given modules Win32 DLL e.g. DESIGN's Wrappers.dll found in the Design subdirectory depends on des.dll. Win32 DLL loading rules apply here. These are:

- The directory from which the importing component loaded. (i.e. the DESIGN subdirectory where Wrappers.dll is loaded in the case of DESIGN)
- The (application's) current directory (where it's run from)
- The Windows directory (default: C:\Windows) (we don't install anything here)
- The directories that are listed in the PATH environment variable

Since the module DLL does not live in the same directory as the Wrappers.dll then unless Marine is running from %PDMSEXEXE% the %PATH% will be used to locate the module DLL. (The Wrappers.dll subdirectory is effectively is the directory which is first looked in for all the Win32 DLL's which Wrappers depends on i.e. <module>.dll, sglnet.dll, udDNet.dll, °)

Make sure that both %PDMSEXEXE% and %PATH% point to the root directory in which Marine is installed.

8.4.3 No Add-ins

Problem: Add-ins missing from module

Solution: Add-ins loaded by each module are defined in xml files, as example below:

```
<application>addins.xml
```

If Add-ins are missing, check that these files exist in %PDMSEXEXE%

8.4.4 Window Layout

Problem: Window layout is not restored e.g. Add-ins position and/or docked state, ...

Solution: The file <module><version>WindowLayout.xml in the %PDMSUSER% directory stores this information for the each GUI module. If this file is deleted then the window layout will return to its default.

8.4.5 Serialization

Serialization is the process used to store the data - e.g. history stacks, toolbar state - for the GUI in a .NET environment.

Problem: Information is stored per project and per module in the following files in the %PDMSUSER% directory:

<project><version>Settings.bin

<module><version>Settings.bin.

Solution: These files may be deleted or transferred between users working on the same project.

8.4.6 Running Diagrams after upgrading Visio2007 to full Office2007

Problem: After upgrading a Visio installation, Diagrams refuses to start giving the following error message in the console window:

'Diagrams has encountered a problem and needs to close. We are sorry for the inconvenience'.

Solution: After upgrading Office/Visio, make sure you always start Visio directly from the Start menu once, allowing it to register itself properly before invoking any dependent application, i.e. Diagrams.

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